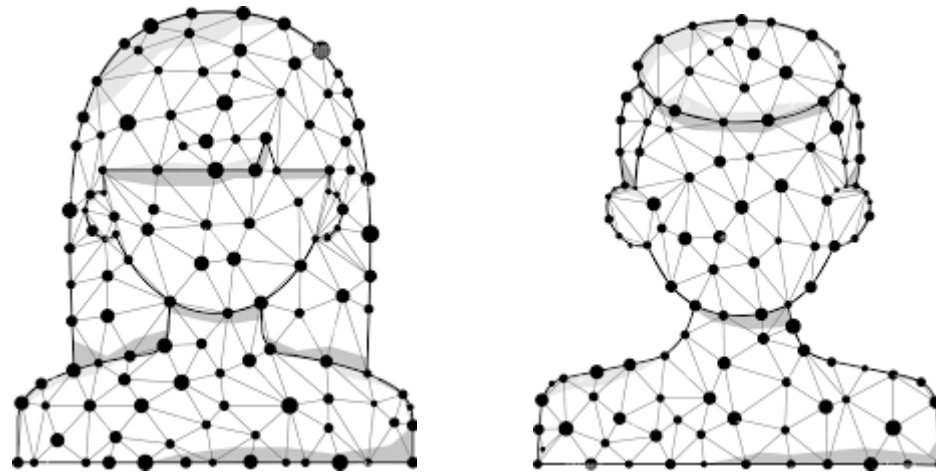




Tecniche e applicazioni biometriche

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Corso di Complementi di biometria



+ Nuovi Argomenti

Lezione 03

Principali tratti monomodali: impronta a contatto

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L3



Impronte digitali a contatto

Nota: le impronte senza contatto
verranno presentate più avanti nel corso



Cosa sono le impronte digitali?

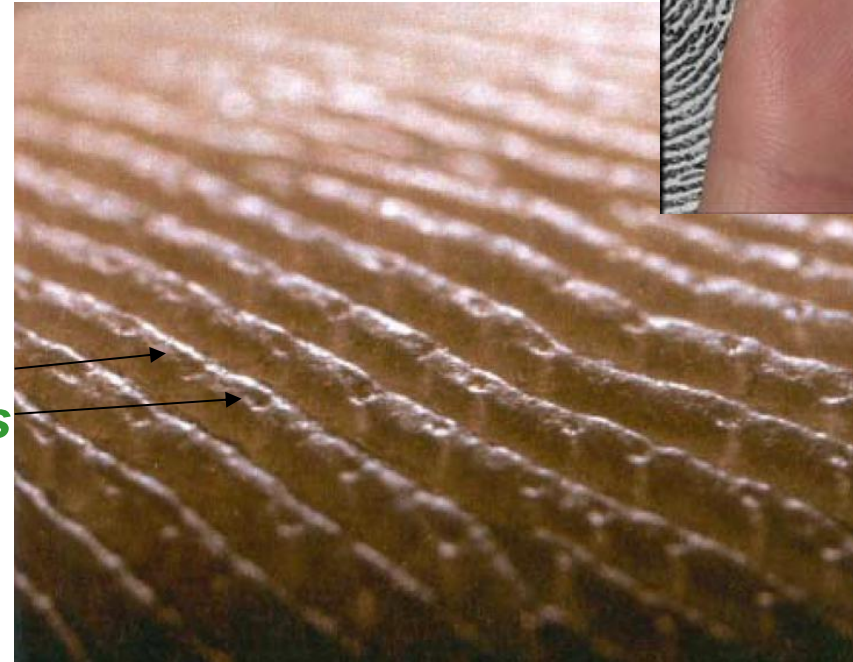
- Sono creste e valli della pelle (Dermatoglyphics) sui palmi e sulle dita di molti animali



Koala Fingerprint



Friction ridges
Pores

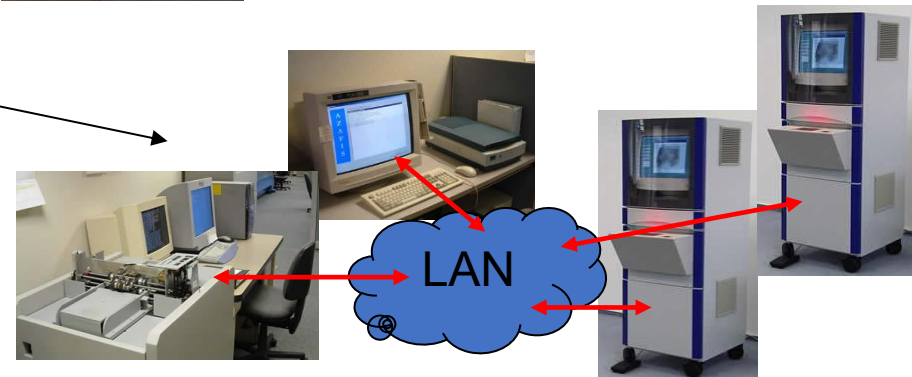
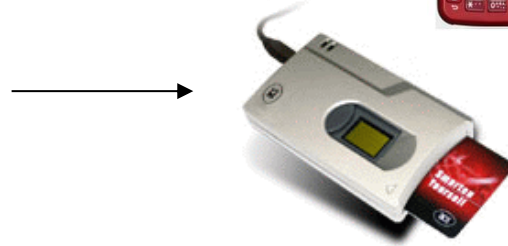
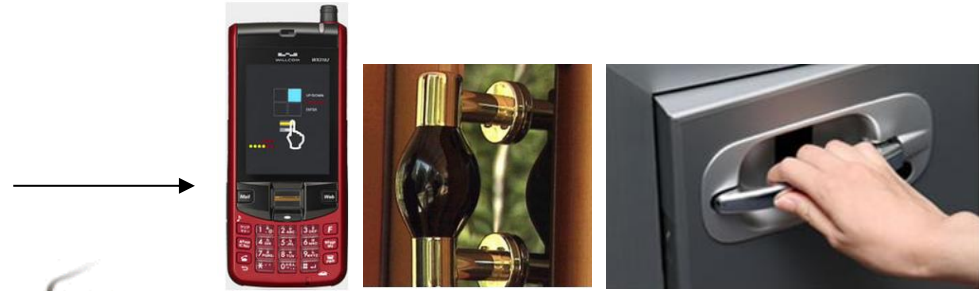


**Sono tratti biometrici stabili
dall'ottavo mese di
gestazione a meno di
abrasioni, malattie o
incidenti gravi**

Diverse scale

Sistemi

- sistemi integrati
- smartcard
- per PC
- “stand alone”
- distributi (AFIS)



Alcune applicazioni basate sulle impronte

<i>Applicazioni forensi</i>	<i>Governative</i>	<i>Commerciali</i>
• Identificazione - dei corpi - delle persone - terroristi • Bambini scomparsi • Attività investigativa	• Carte d'identità e passaporti • Patenti di guida • Controllo accessi • Controllo delle frontiere • Controllo documenti	• ATM (bancomat,..) • Logon su terminali • Ecommerce • Accesso ad servizi online • Telefoni cellulari • PDA • Controllo accessi

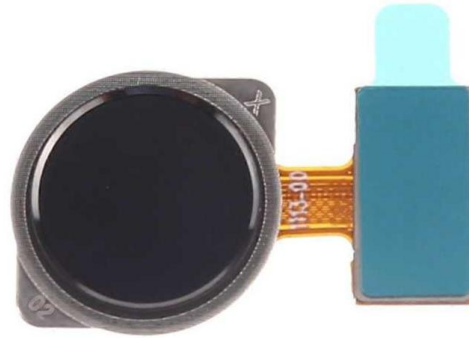
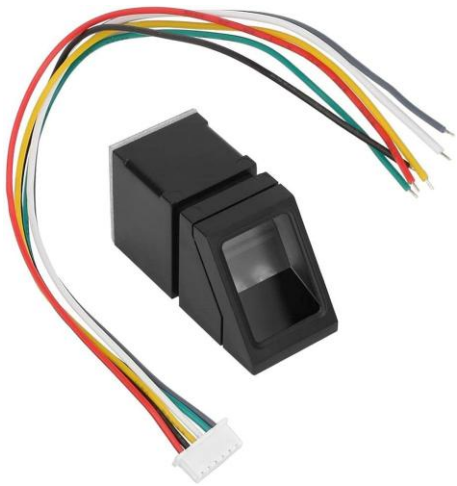


Punti di forza e debolezze

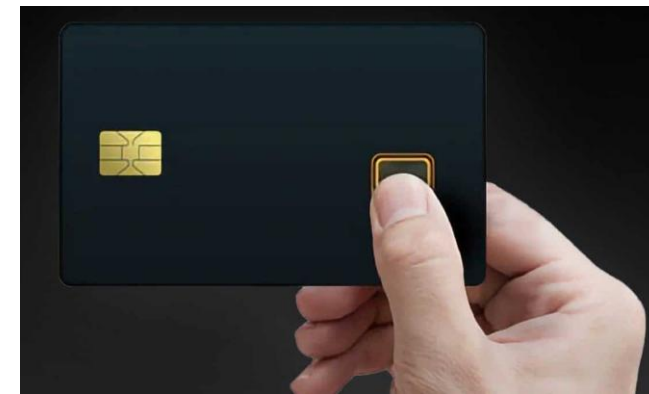


- Punti di forza
 - E' una tecnologia matura, ampiamente controllata, altamente accurata e funzionante in molti ambienti operativi
 - L'acquisizione è mediamente facile ed ergonomica
 - Offre la possibilità di usare più dita, aumentando notevolmente l'accuratezza dei sistemi
- Debolezze:
 - Tutti i dispositivi non riescono ad acquisire le impronte di una frazione della popolazione (solitamente il 4%)
 - L'accuratezza tende a degradare nel tempo
 - Essendo largamente associata ad applicazioni forensi, alcune persone provano senso di disagio nel fornire il tratto biometrico





SENSORI e SISTEMI COMMERCIALI PER L'IMPRONTA A CONTATTO

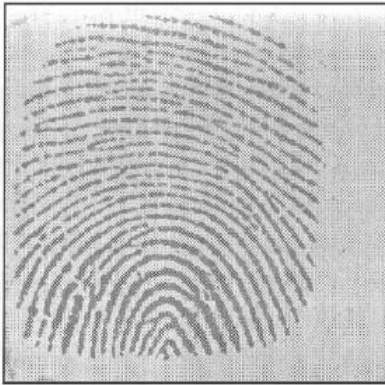


Panoramica delle immagini dei sensori

I vari sensori producono delle immagini con delle caratteristiche molto diverse fra loro



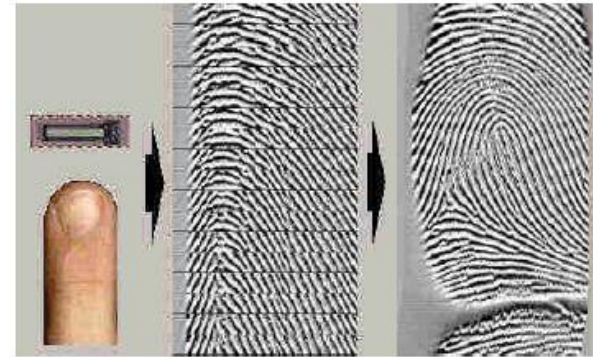
Optical scanner



Capacitive scanner



Piezoelectric scanner



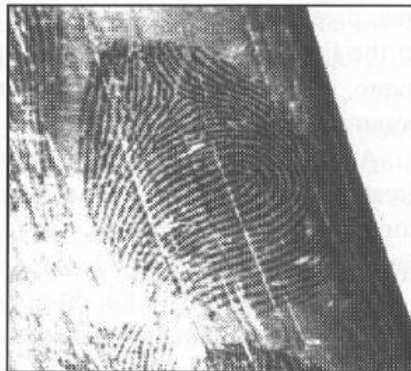
Thermal (Sweeping)



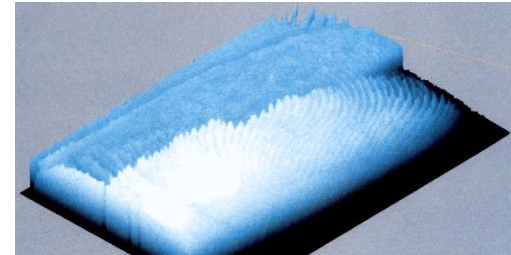
Thermal scanner



Inked impression



Latent fingerprint

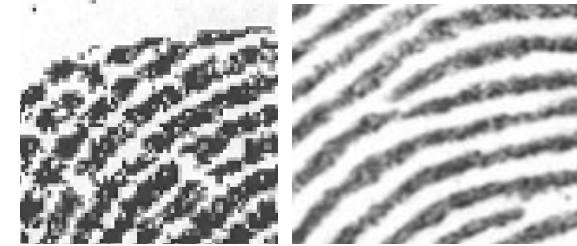


3dimensional

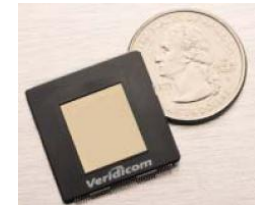
Proprietà del sensore

- Nello scegliere un sensore dobbiamo controllare:

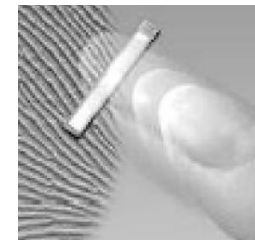
- risoluzione



- area di acquisizione



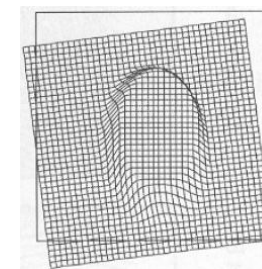
- numero di pixel e bit per pixel (8, 16, 24 bit)



- contrasto



- distorsione geometrica

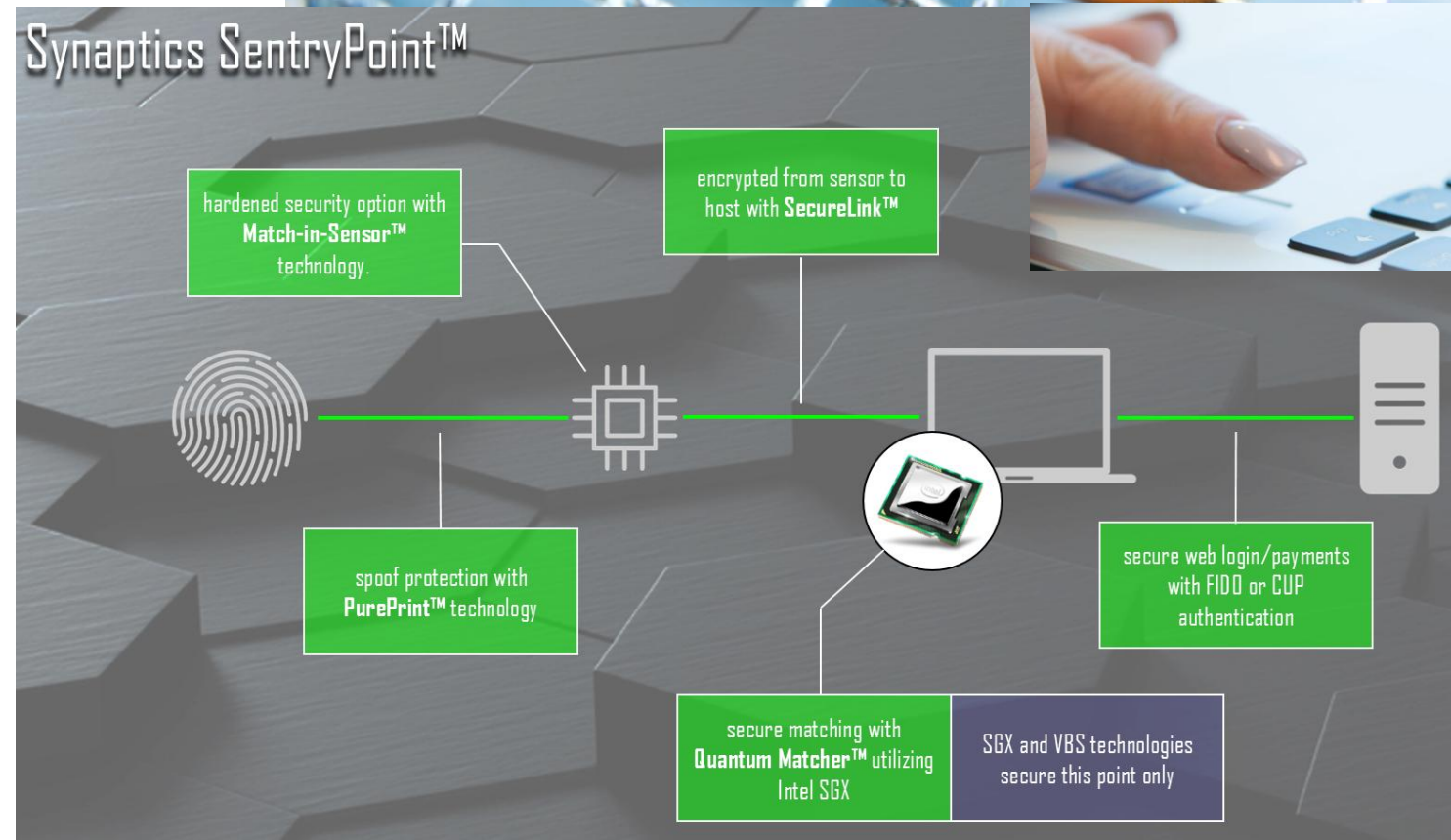
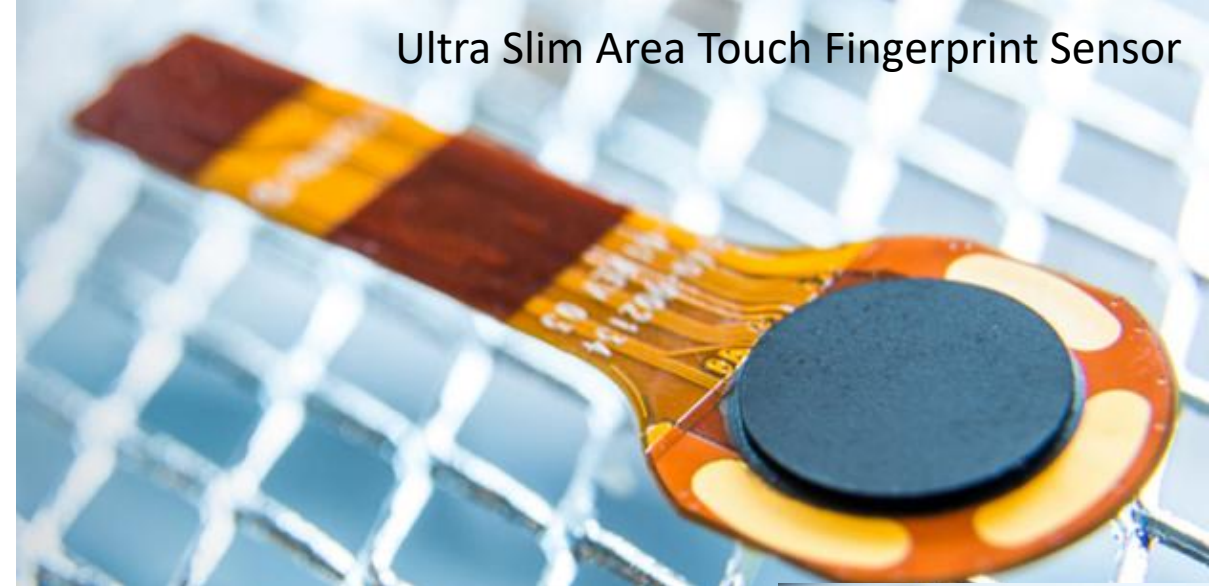


Smartphone Fingerprint Sensor

Synaptics SentryPoint

- Encryption
 - All biometric data is encrypted and digitally signed using strong 256-bit Advanced Encryption Standard and Transport Layer Security technologies to prevent eavesdropping, tampering or fraud.
- Match-in-Sensor architecture
- Anti-spoof technology

Ultra Slim Area Touch Fingerprint Sensor



Under

(capacitivo)



- Synaptics FS4500: 6x6mm Natural ID™ fingerprint sensor solution authentication through **capacitive** touch fingerprint sensing through
 - Glass
 - Ceramic buttons,
 - Recessed cover glass up to 300um thick.
- 300+ million Natural ID (capacitive) units
 - Lenovo, Huawei, ...

Tecnologia “**Under the glass**”
Permette di rendere il sensore
ancora meno invasivo e “immerso”

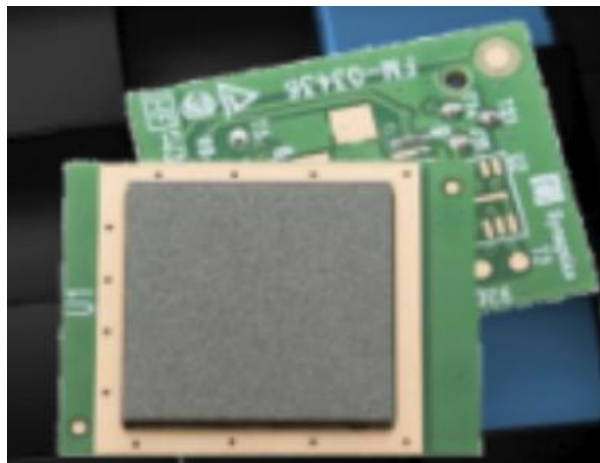
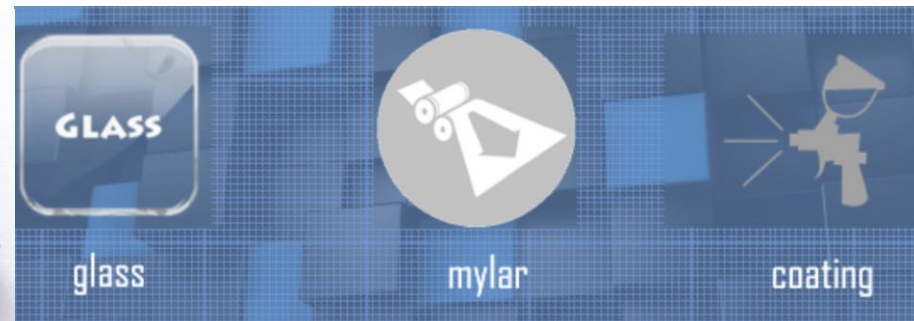


Match-in-sensor: Synaptic FS7600 (capacitive)

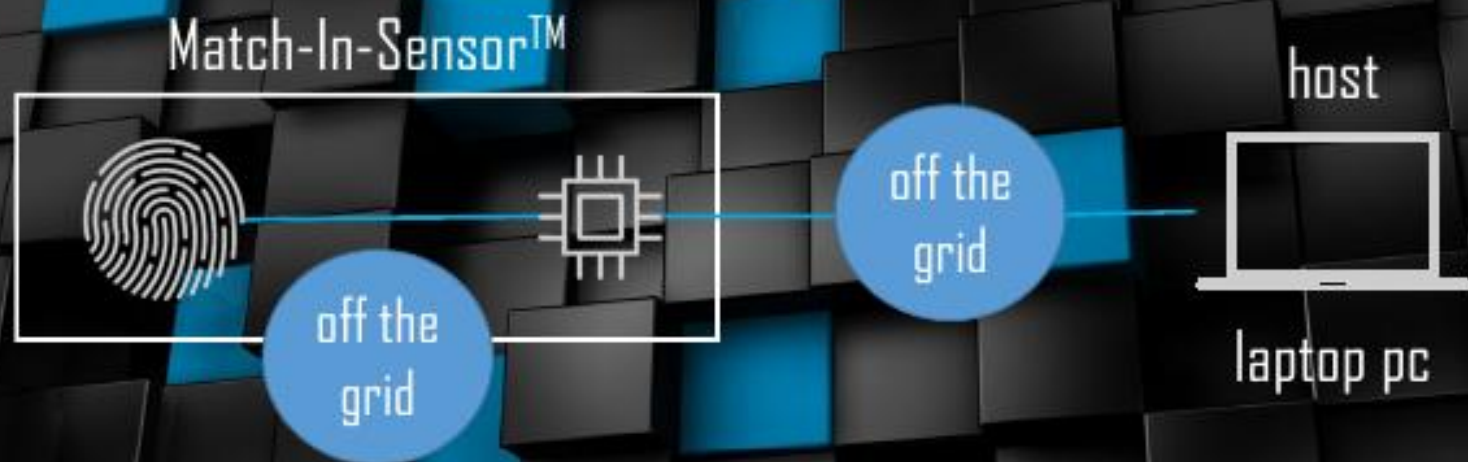
(FS7600 supports up to a 0.2 mm sensing distance, meaning it can be put under glass, under mylar, or just coated with a protective layer)



Xiaomi, PQI



FS7600 highest security for user fingerprints

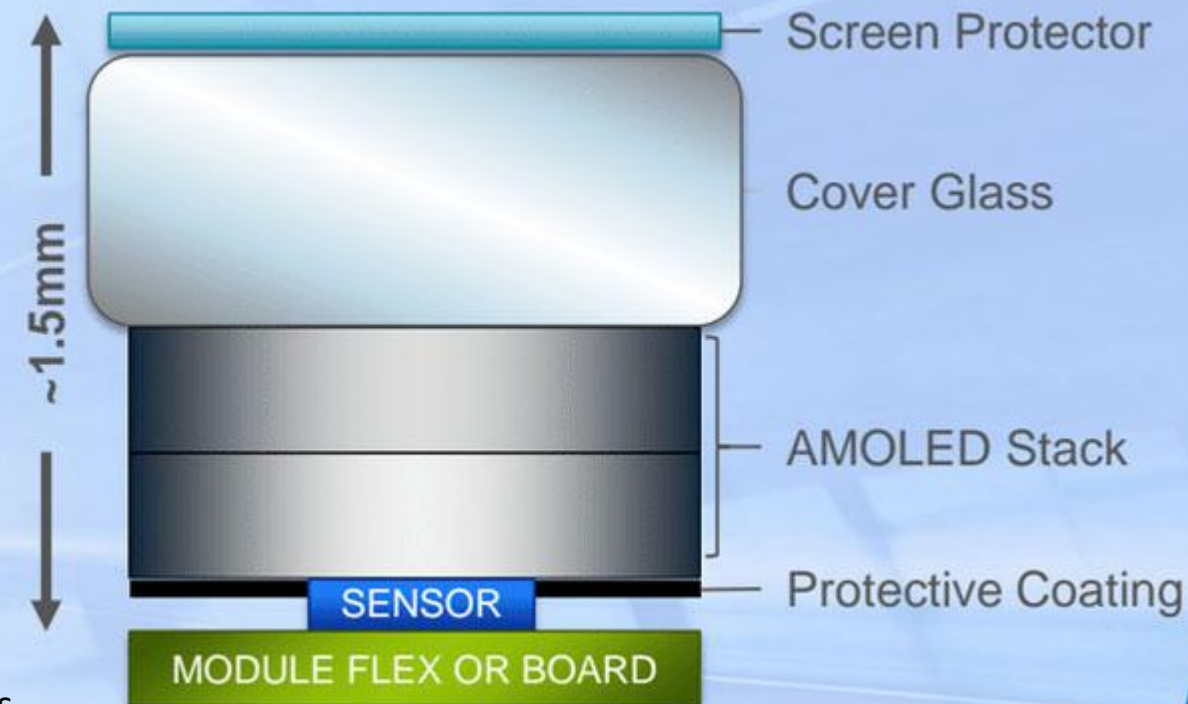
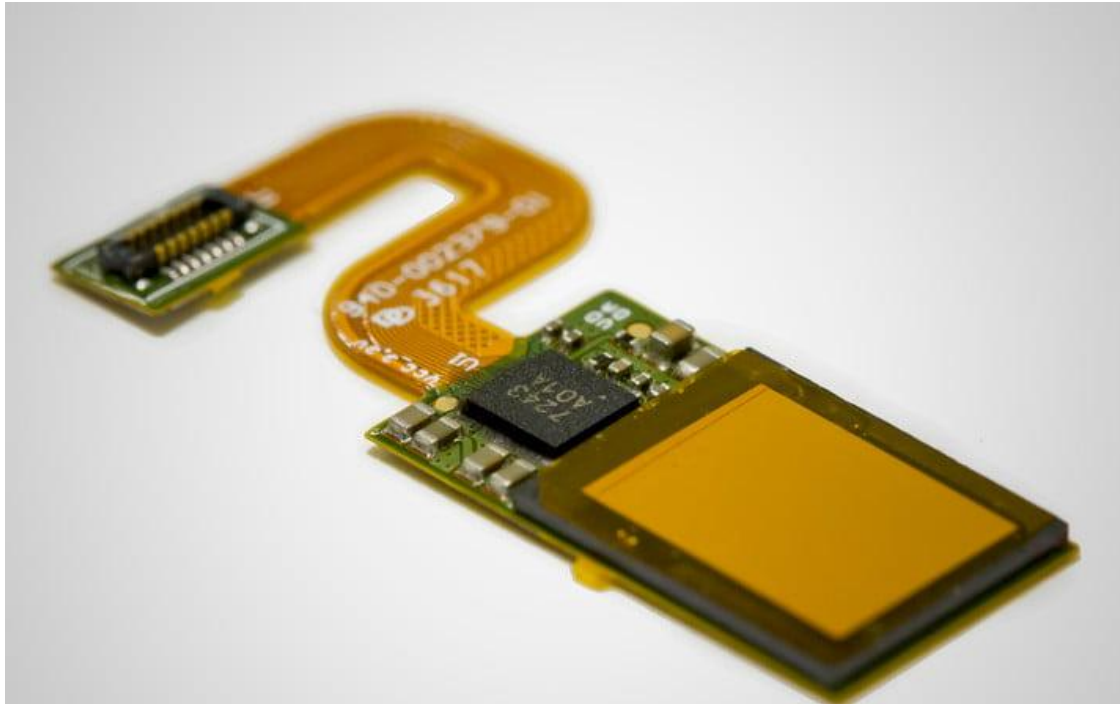


FS7600 "off the grid" – non accessible by OS

fully hardware encapsulated fingerprint database and matching solution

The Synaptics Clear ID FS9500

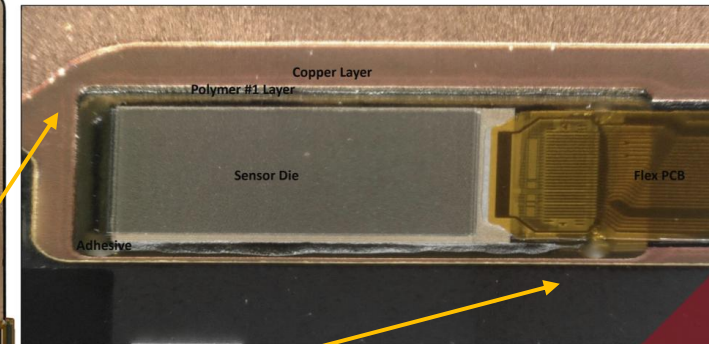
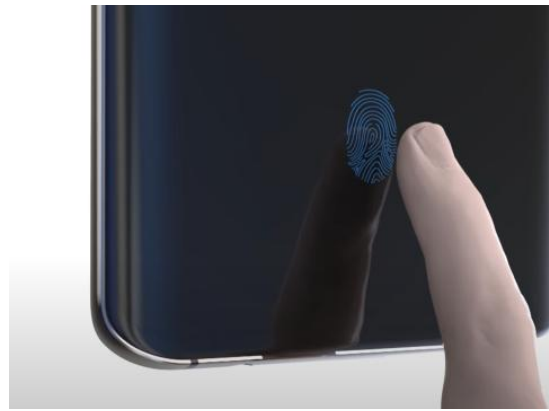
- **Optical** in-display fingerprint sensors
- Works with wet, dry and cold fingers



Interfacciabile facilmente con Arduino, Raspberry PI and other microcontrollers

Qualcomm 3D Sonic Sensors

- First **ultrasonic** fingerprint readers on mobile devices



Qualcomm 3D Sonic Sensor Assembly
©2019 by System Plus Consulting

SAMPLE



Qualcomm 3D Sonic Max

Qualcomm[®]
3D Sonic Max
World's largest fingerprint sensor

Advanced security features
Simultaneous two-finger sensing
17x bigger
1::1,000,000 vs. 1::50,000 accuracy

Press fingerprint icon to enroll



Resistenza ai contaminanti classici

Avere due impronte apre moltissimi scenari
(multimodale, classificazione, antispoofting,...)



Esempi di sensori/sistemi commerciali per sicurezza fisica di alta gamma (high end)

FVC-onGoing competition 2010:
Suprema's 'SFCore' algorithm **EER 0.258%**

INCITS (*):
FNMR=0.23% @ FAR=0.01

Fingerprint Matching Benchmark:
FMISO-STD-1.0 (examines matching performance of fingerprint algorithm in ISO19794-2 template for interoperability of biometric systems) Minutiae Interoperability Exchange (MINEX) test by the NIST, 2008
Evaluation of the INCITS 378 fingerprint template

FVC-onGoing:
EER 0.258%

Nel 2025 i migliori sistemi hanno
EER <0.06%

False Non-Match rates at a fixed False Match Rate of .01
measured over 125,000 genuine comparisons:

Suprema's 'matcher' algorithm showed a 0.23% false rate (media EER), 'Extractor' algorithm recorded a 0.30% false rate

Sono arrivati sensori più recenti sul mercato, riferirsi a

- IDEMIA linea SIGMA
- Bosch Suprema BioEntry W2



Esempio di sistema commerciale

IDEMIA linea SIGMA

Idemia ha
installato 5M
di dispositivi



SIGMA Lite

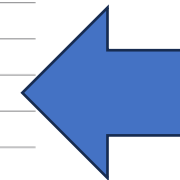
SIGMA Lite+

SIGMA Wide

SIGMA Extreme

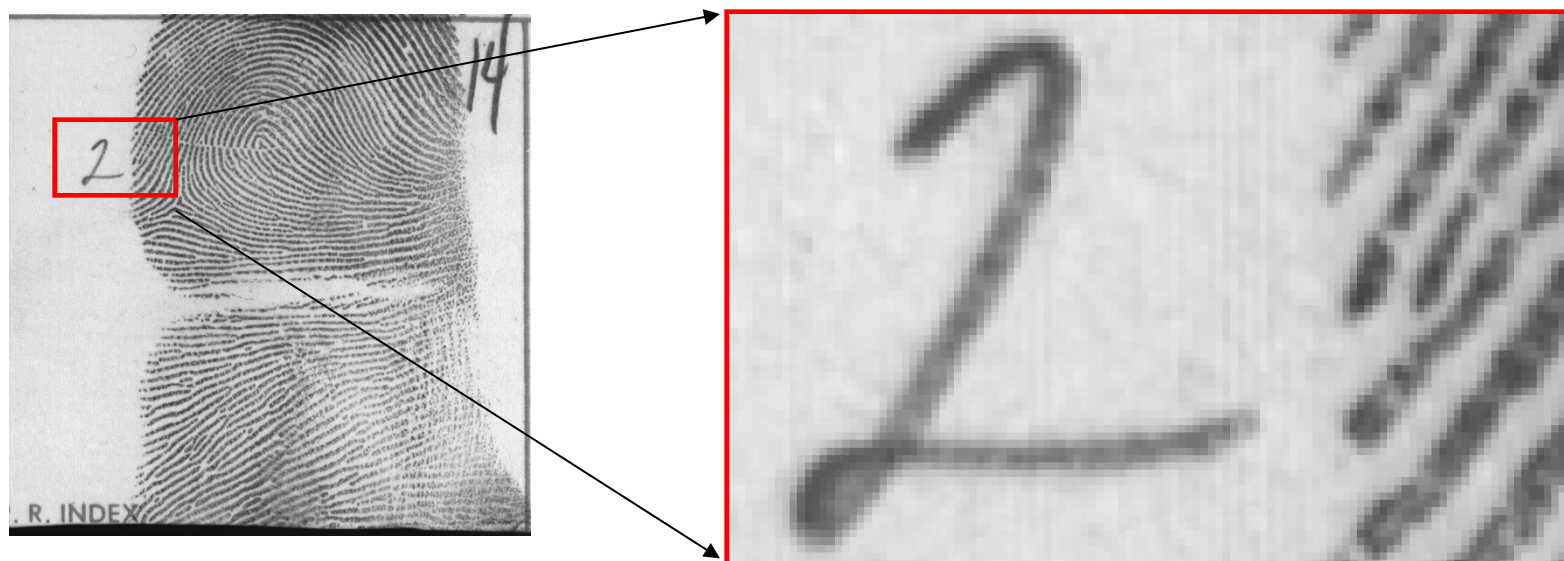
Fingerprint capability				
Sensor size	14x22mm			23x23mm
User records	500, extendable to 3K or 10K		5K, extendable to 10K, 50K or 100K	
Duress finger prevention	✓	✓	✓	✓
Fake finger detection	Statistical images comparison			Patented electro-optical tech.
IDs in authorized user list	250 K			
Tansaction log capacity	1 million			
Model variants for contactless cards				
No card reader	✓	-	-	-
HID® Prox	✓	✓	-	✓
HID® iClass® Legacy SE/SEOS	✓	✓	✓	✓
MIFARE® & MIFARE DESFire®	✓	✓	✓	✓
User interactivity				
Screen	-	2.8"	5" horizontal	5" horizontal
Audio	Buzzer	Buzzer	Loudspeaker	Loudspeaker
video	-	-	✓	✓
Time & attendance	-	Basic	Full	Full

- SIGMA Extreme
- Sensore 23x23mm
- Fake detection elettrica
- 250.000 utenti registrati
- Card reader (iClas MIFARE, sono tecnologie per crittazione per RFID)
- -20°C to 60°C
- IP65 (+-ok pioggia)
- Ethernet, USB, WIFI
- 12-14V PoE



Immagini delle impronte

- Il sample della impronta è una immagine in toni di grigio quindi si controlla
 - risoluzione,
 - bit per pixel
- Un esempio: l' FBI digitalizza le impronte del DB nazionale a 500 Dpi con 8 bit per pixel. Un cartella con 10 impronte occupa circa 10 MB!



Spoofing e controlli

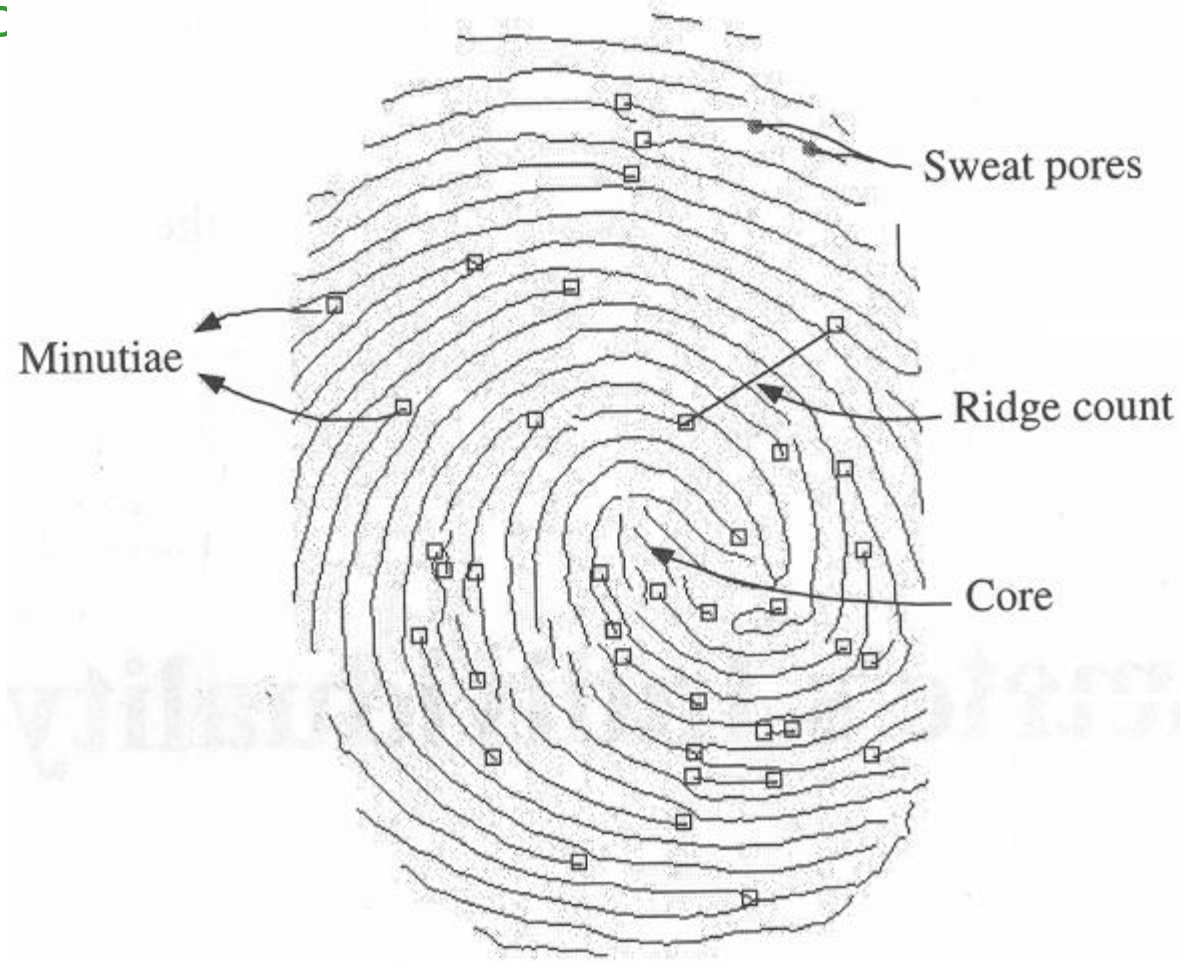
Confrontiamo le immagini ottenute dei sensori

	Dito reale	Dito in silicone	Dito in gelatina
Sensore Ottico			
Sensore Capacitivo		Il sensore capacitivo non riesce a rilevare bene il dito in silicone! (auto-difesa!)	

UNICITA CLASSIFICAZIONE DELLE IMPRONTE

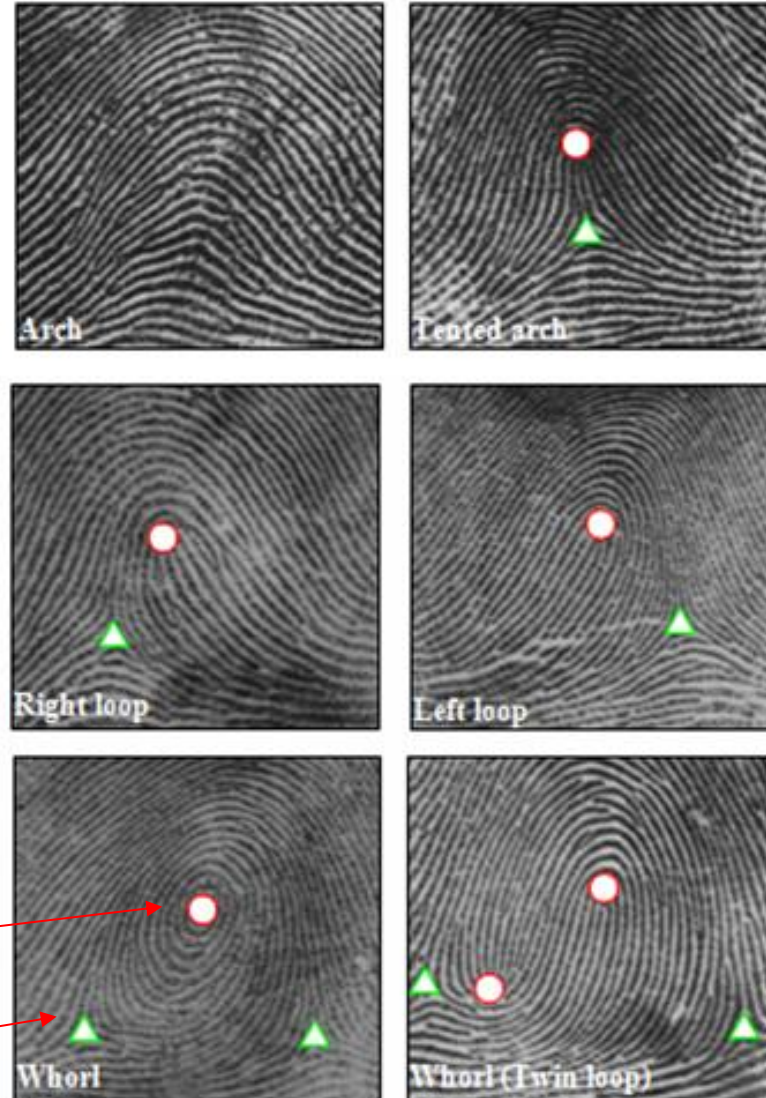
Unicità delle impronte (4)

- In realtà abbiamo a disposizione **molti altri parametri** quindi la stima diventa ancora **più cc**



Sistema di classificazione attuale (1)

- Le impronte si dividono in
 - Arch (Plain e Tented)
 - Loop (Left, Right)
 - Whorl (Plain, Twin loop)
- attraverso
 - l'individuazione degli eventuali Core e Delta
 - lo studio degli orientamenti dei ridge

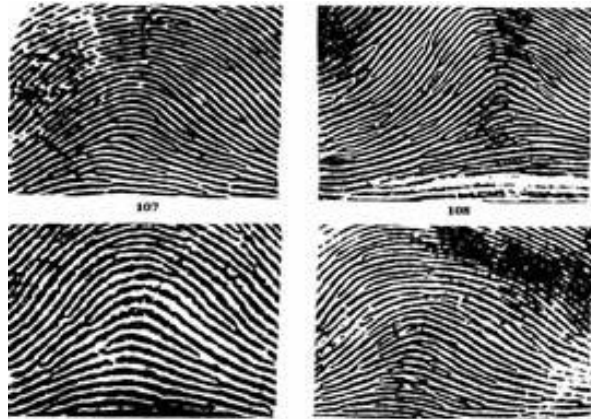
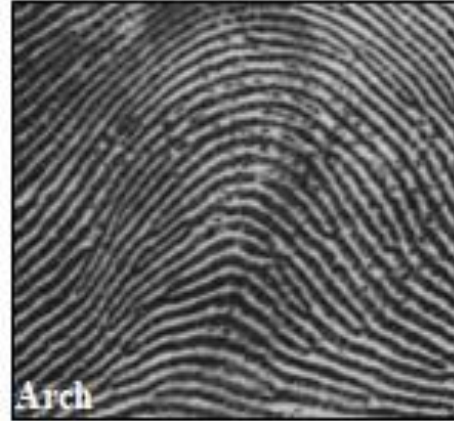


Core

Delta

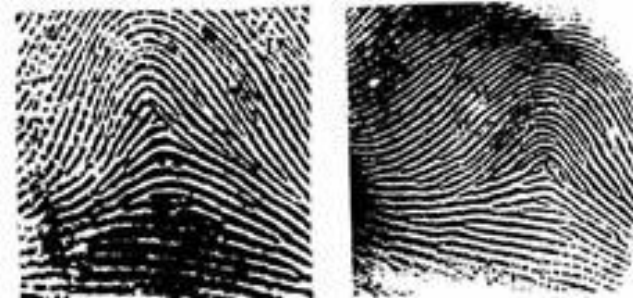
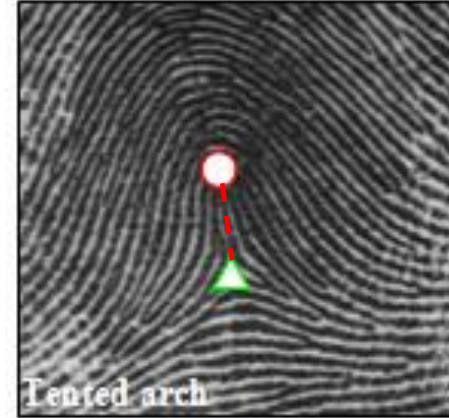
Arches

Plain arch



Ridges enter on one side of the impression and flow or tend to flow out the other with a rise or wave in the center.

Tented arch

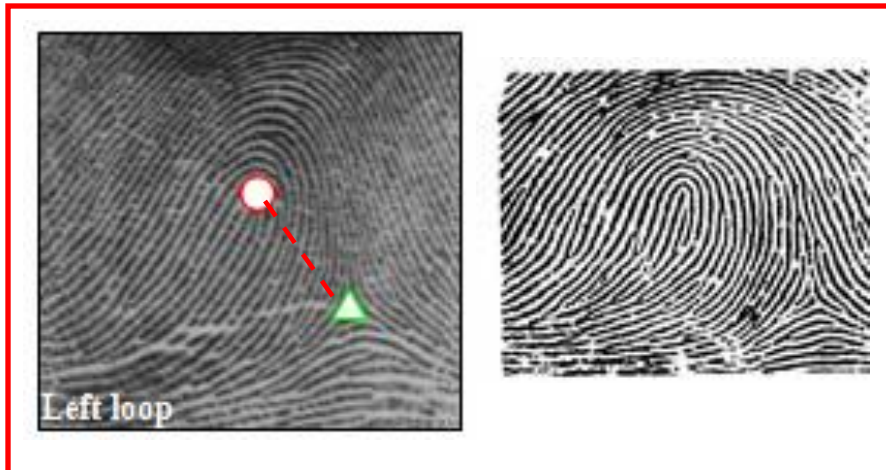


Most of the ridges enter upon one side of the impression and flow or tend to flow out upon the other side, however the ridges at the center do not. Tend to form an upthrust or angle at the center of the pattern

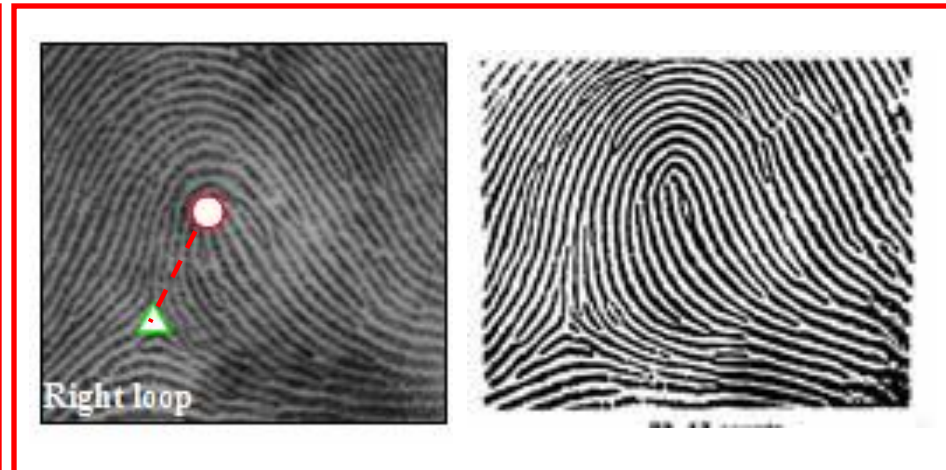
Loops

- A loop is that type of fingerprint pattern in which one or more of the ridges enter on either side of the impression, recurve, touch or pass an imaginary line drawn from the delta to the core, and terminate or tend to terminate on or toward the same side of the impression from whence such ridge or ridges entered.
- Essentials of a loop:
 - Sufficient recurve;
 - One delta;
 - A ridge count across a looping ridge.

Left loop



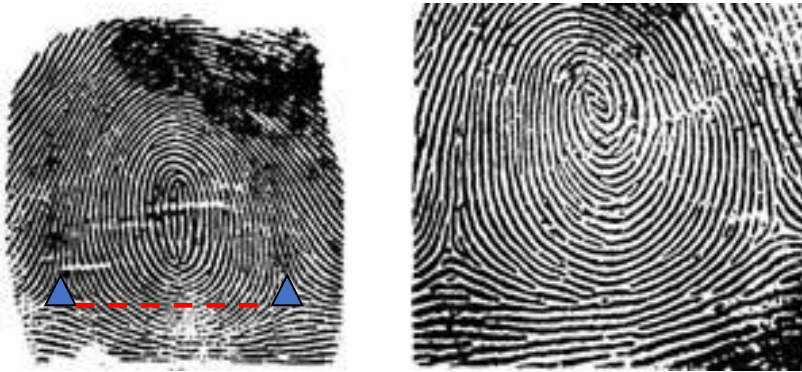
Right loop



Whorls ⁽¹⁾

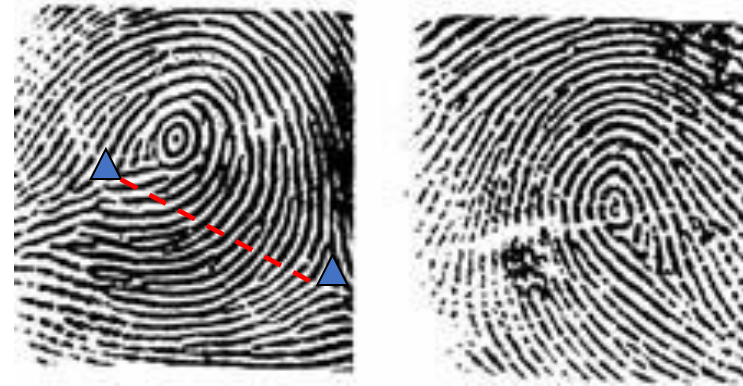
- ***at least two*** deltas are present
- with a recurve in front of each

Plain whorl



Has two deltas and at least one ridge making a complete circuit, which may be spiral, oval, circular or any variant of a circle. An imaginary line drawn between the two deltas must touch recurving ridge in the pattern area.

The Central Pocket Loop



Has two deltas and at least one ridge which makes a *complete circuit*. A line drawn between the two deltas must not touch or cross any recurving ridges within the inner pattern area

Whorls (2)

The Double Loop



Two separate loop formations, with two separate and distinct sets of shoulders and two deltas

Accidental

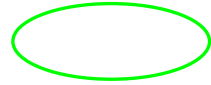


Combination of two different types of patterns, with the exception of the plain arch, with two or more deltas; or a pattern which possesses some of the requirements for two or more different types or a pattern which conforms to none of the pattern definitions.

Tre livelli di analisi

- Una impronta può essere esaminata su tre livelli:

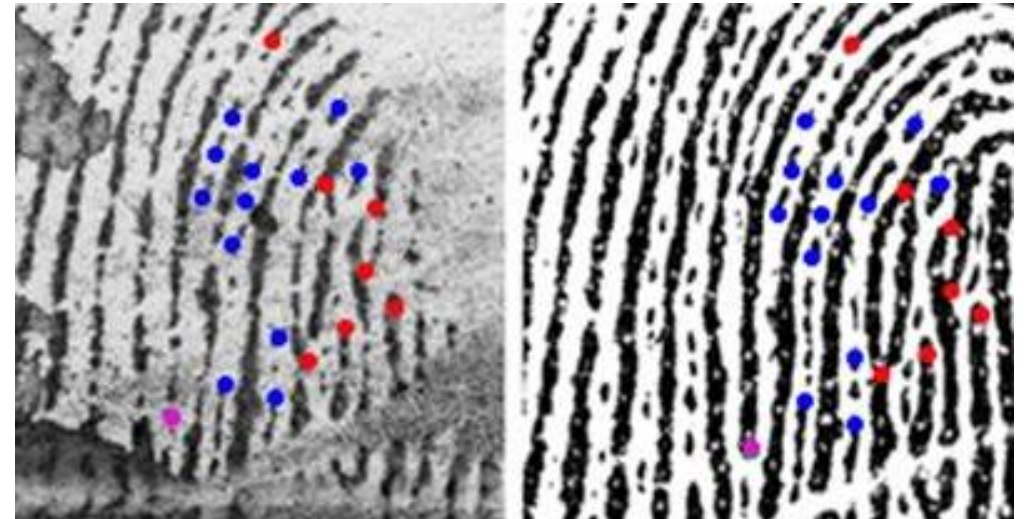
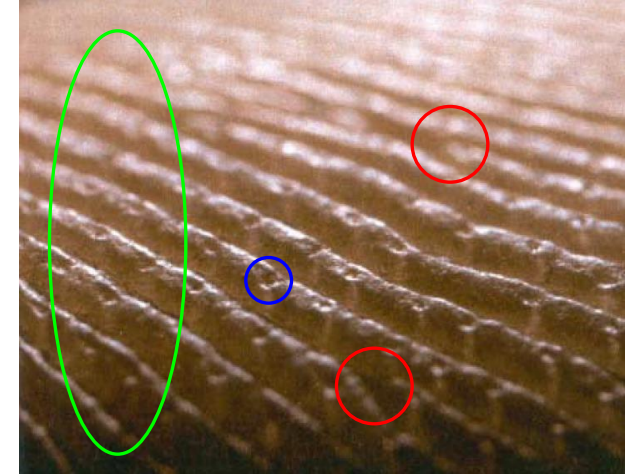
- globale (livello I)



- locale (livello II)



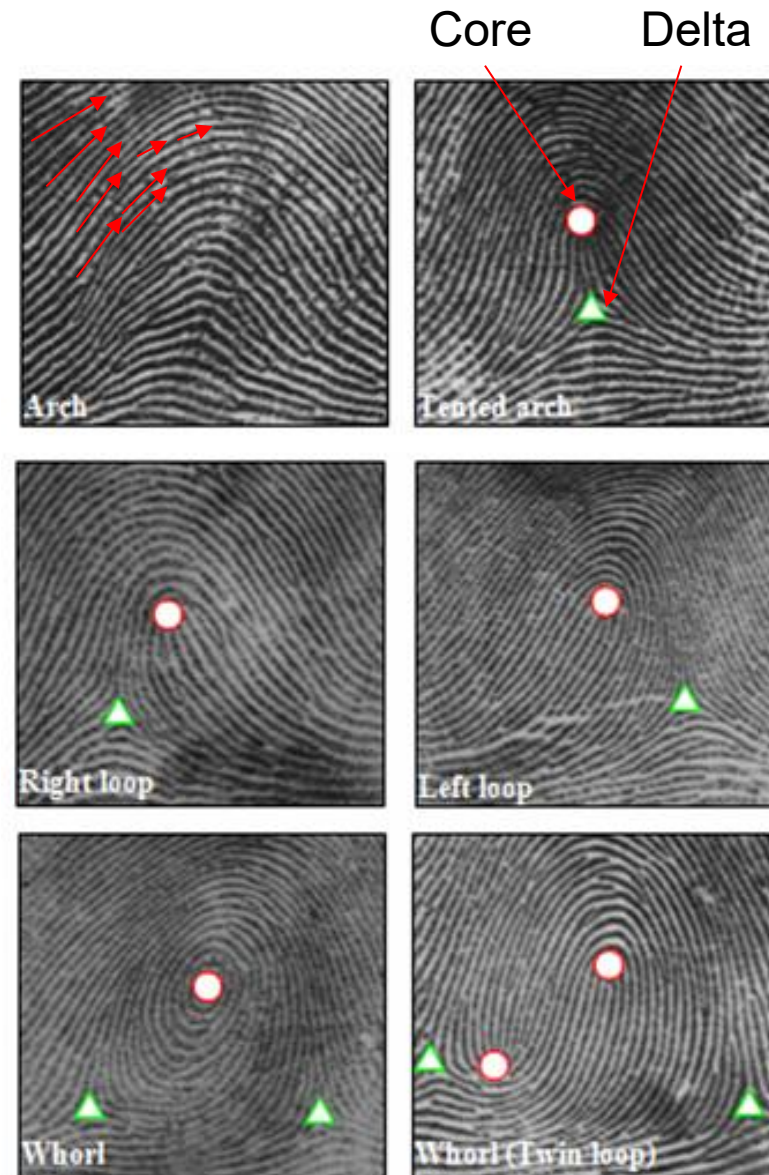
- ultra-fine (livello III)



Livello I

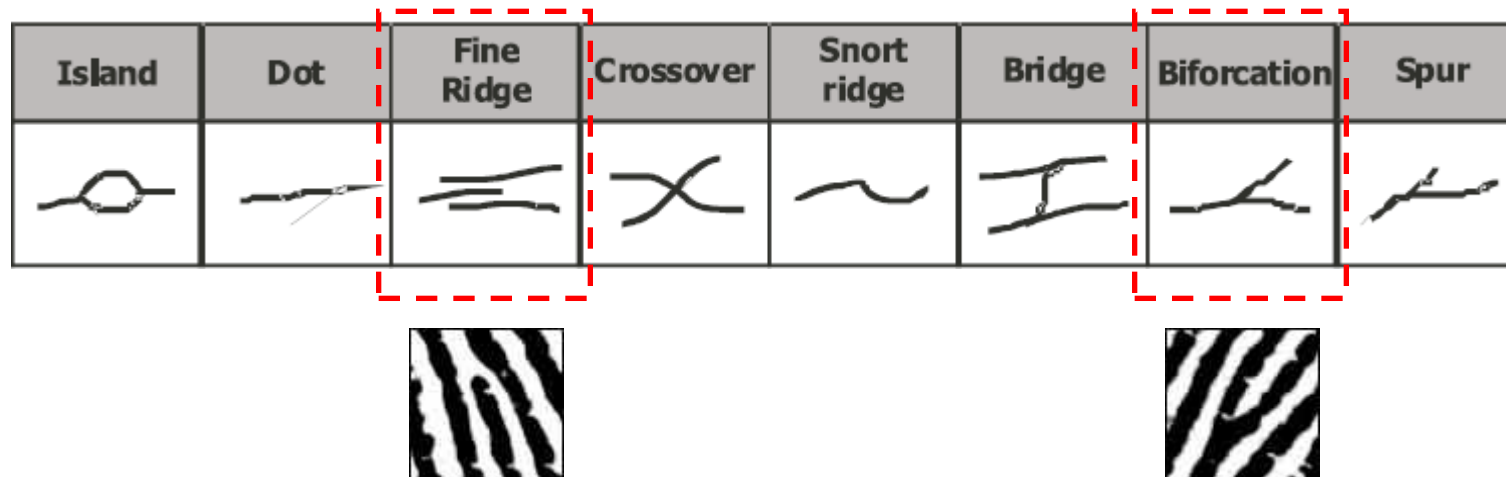
A livello globale si osservano:

1. il flusso delle linee (arch, loop, whorl o sottoclassificazioni)
2. i punti singolari (core, delta), accurato.
3. la forma dell'impronta
4. l'orientamento
5. la frequenza delle righe dell'immagine.



Livello II

- A livello locale è possibile identificare fino a circa 150 diverse caratteristiche locali delle creste (minutiae)



- Le due principali caratteristiche sono le terminazioni e le biforcazioni
- L'FBI per i sistemi automatici (AFIS) usa solo terminazioni e biforcazioni

Livello III

A livello ultra-fine è possibile individuare i seguenti dettagli:

- intra-creste (i pori per la sudorazione)
- inter-creste (incipient ridges)



Pori
(forma e
numero)



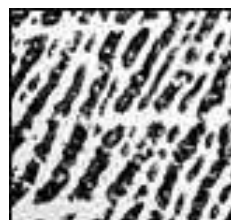
Variazioni
di forma
delle linee



Cicatrici
(permanenti)



Incipient
ridges



Pieghe
(permanenti)

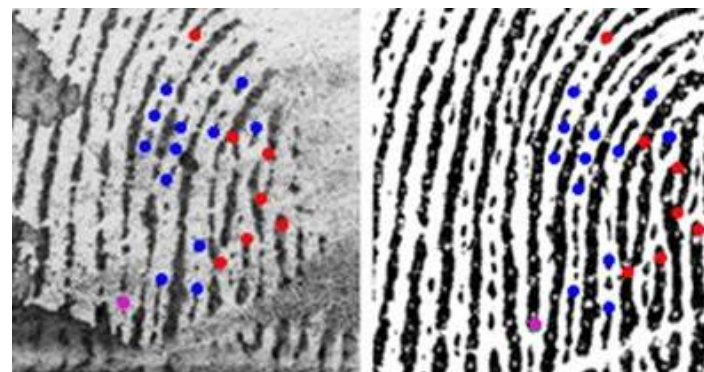


Deformazioni



Verruche
(non
permanenti)

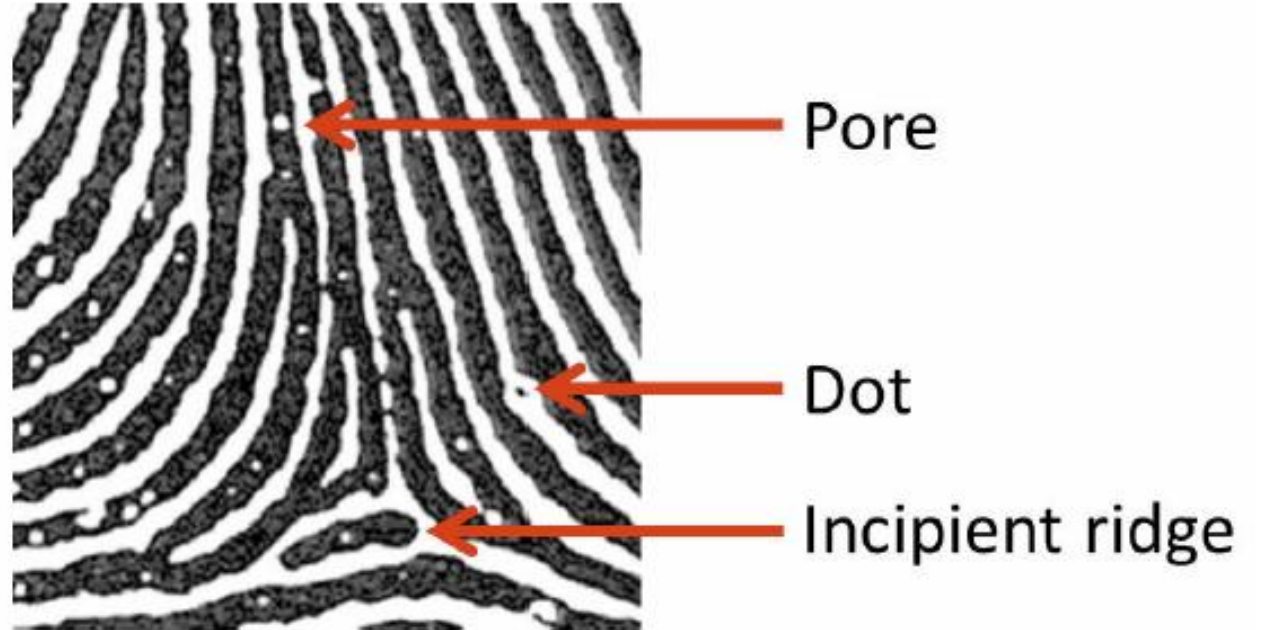
I dettagli del livello III sono considerati altamente distintivi, ma si rilevano solo ad altissima risoluzione, almeno 1000 dpi ed in condizioni ideali (i pori hanno dimensione 60-250 μm)



Esempi al Livello III

Esempio a 800 DPI

(si vedono solo alcuni pori,
anche se dipende dalla pressione)



Esempio a 1000 DPI

(si vedono i pori)



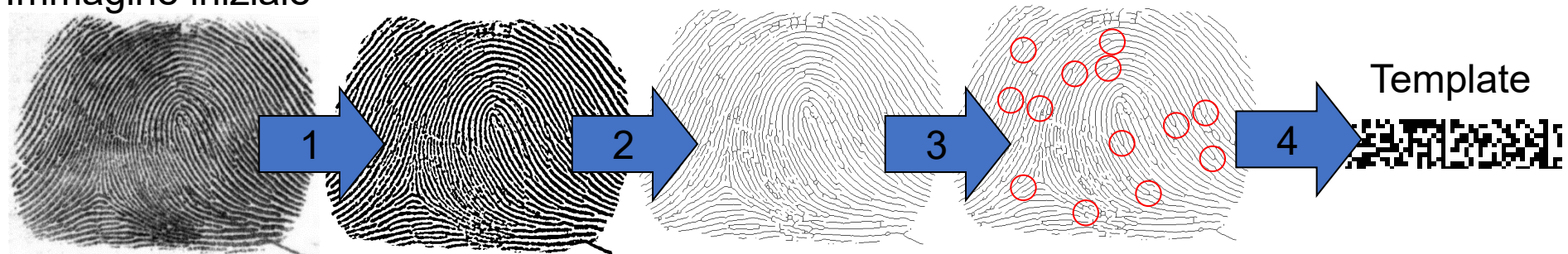
ELABORAZIONE DELLE IMPRONTE

Sequenza di passi tipica

Nel modulo per **l'Estrazione delle feature** si eseguono tipicamente questi passi:

1. Filtraggio iniziale
2. Manipolazione della immagine (enhancement)
3. Estrazione delle feature
4. Codifica (*)

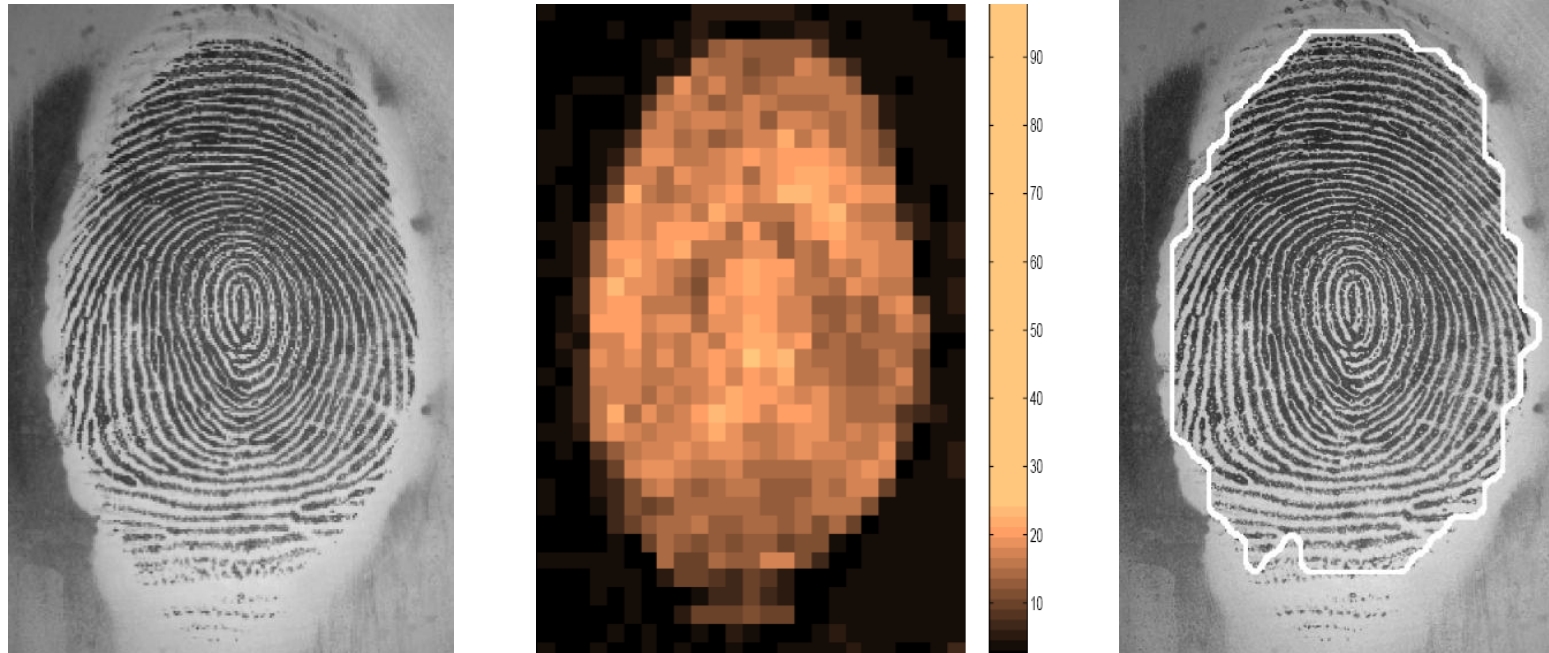
Immagine iniziale



(*) Il modulo di codifica è talvolta rappresentato esternamente al modulo di feature extraction

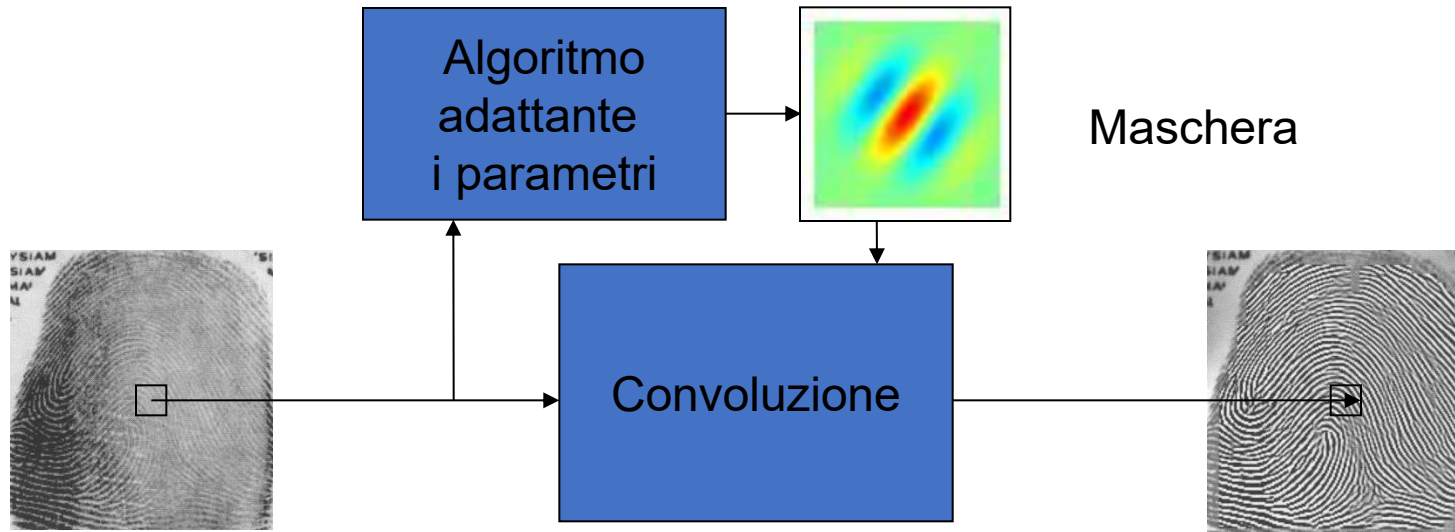
Segmentazione

- Gli algoritmi per la segmentazione estraggono il foreground (l'impronta) dal background (lo sfondo)
- Permettono di focalizzarsi solo sulle regioni della immagine che portano informazione utile al processo biometrico
- Una tecnica basilare molto usata calcola la varianza locale dell'immagine (oppure il modulo del gradiente dell'immagine) in blocchi (16x16) e scarta i blocchi sotto ad una soglia prefissata

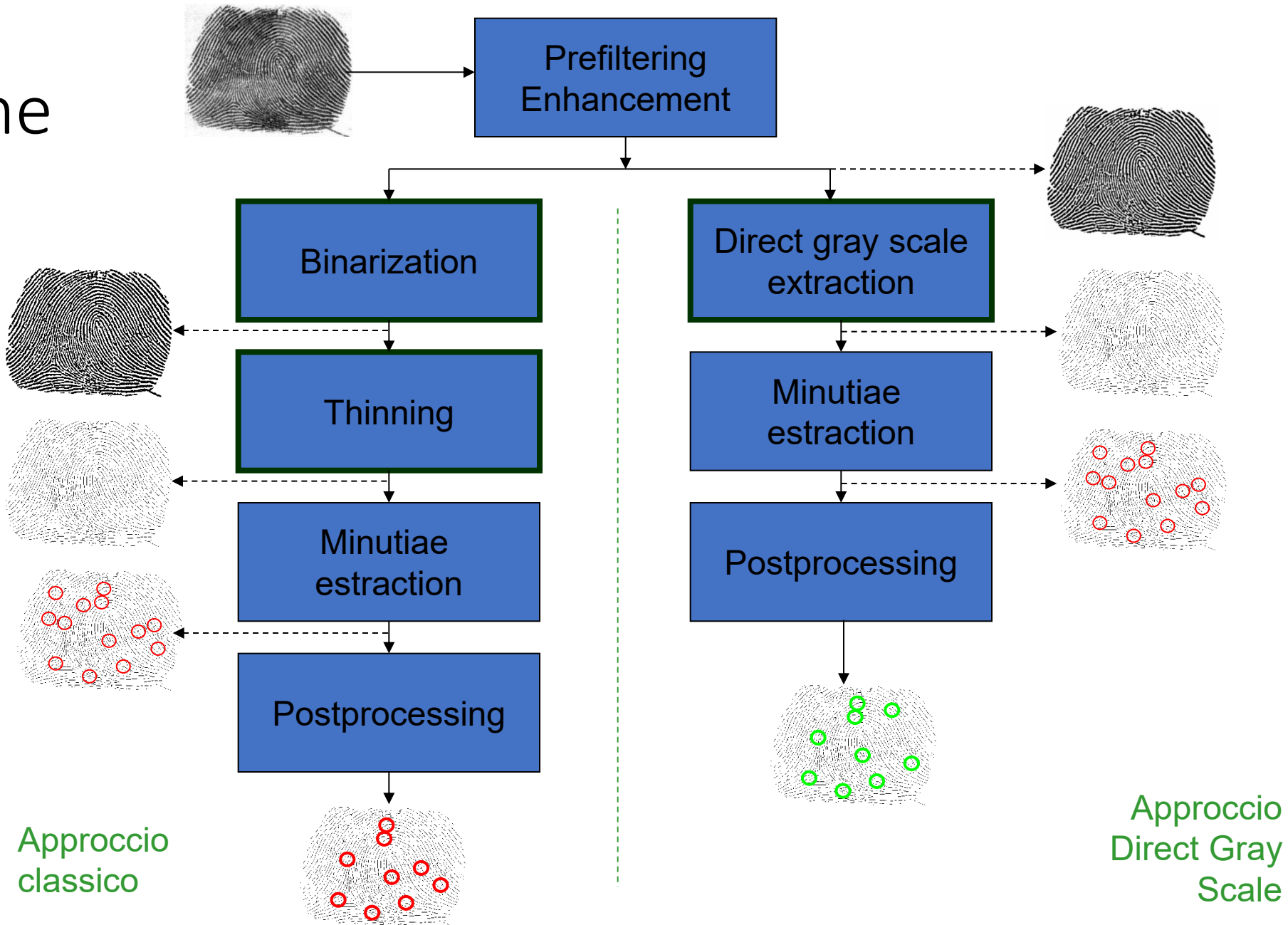


Manipolazione delle immagini: filtri contestuali (2)

- Questi filtri lavorano sulla immagine in ingresso attraverso un'operazione chiamata convoluzione con una maschera di filtraggio
- A seconda del tipo di maschera usata il filtro aumenta/diminuisce alcune caratteristiche piuttosto che altre
- Una parte del filtro controlla la porzione dell'immagine in esame e adatta i parametri della maschera



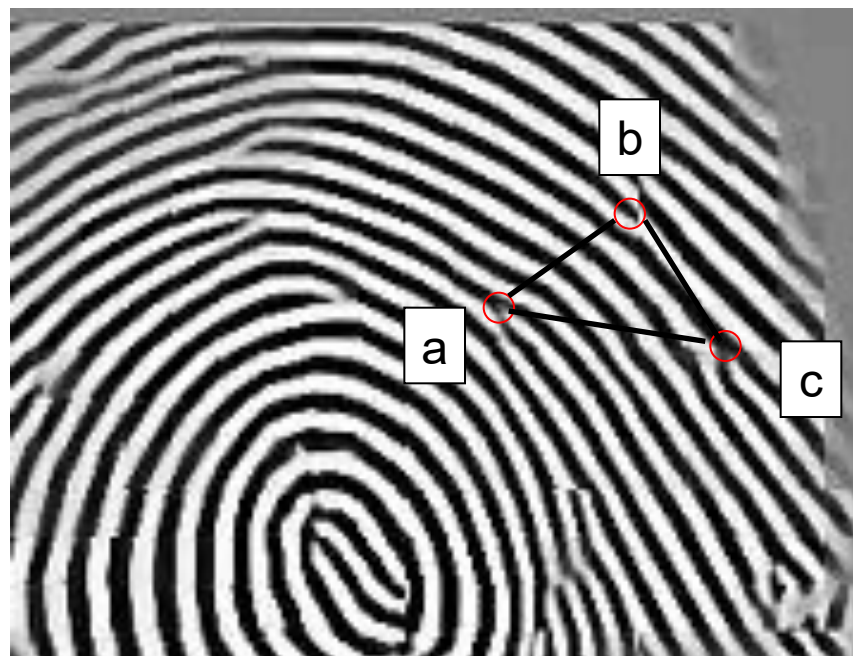
Panoramica delle tecniche



Livello 1: ridge counting

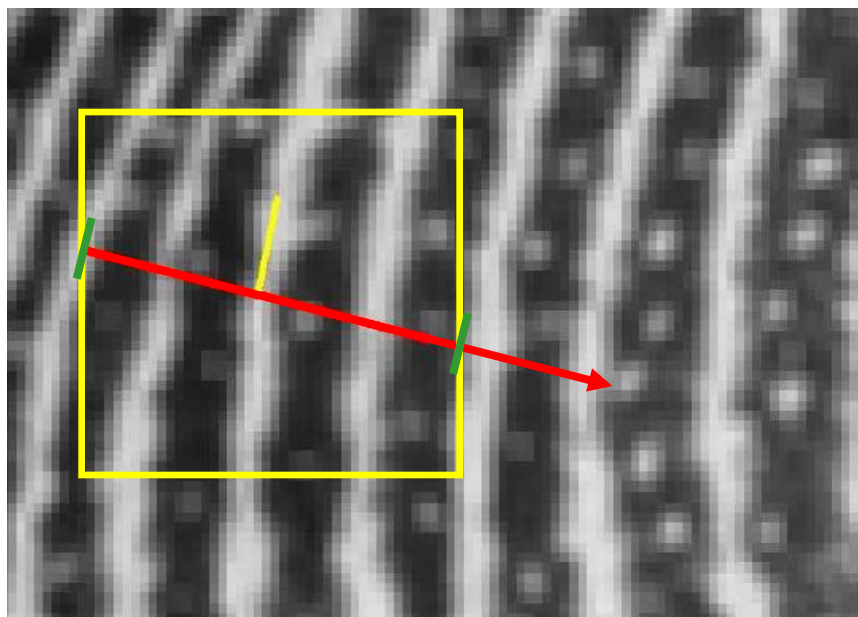
- E' una misura dei ridge che attraversano una linea immaginaria passante tra due minutiae
- In questo esempio abbiamo:
 - fra *a* e *b* 4 ridge
 - fra *b* e *c* 0 ridge
 - fra *c* e *a* 3 ridge

N.B. La grandezza del ridge counting, pur essendo calcolata partendo dalle minutiae (livello II) è considerata come **appartenente al livello I** in quanto porta informazioni non localizzate attorno ad un punto ma su cosa succede nell'impronta fra alcuni punti anche lontani fra loro

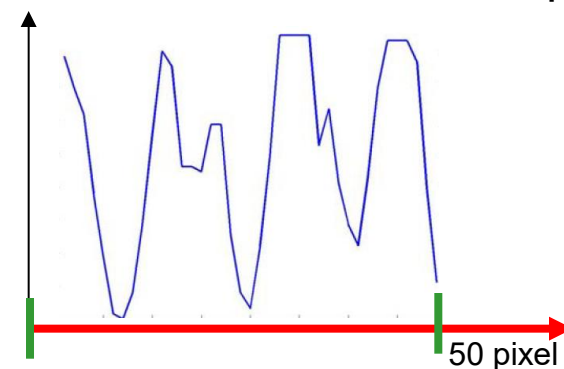


Livelli 1: analisi delle frequenze spaziali

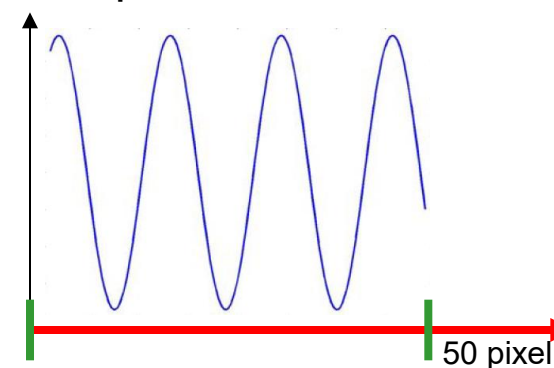
- E' una misura di quanto sono stretti o larghi i ridge nelle varie regioni dell'impronta
- **Ridge frequency**: inverso della distanza media tra due picchi consecutivi



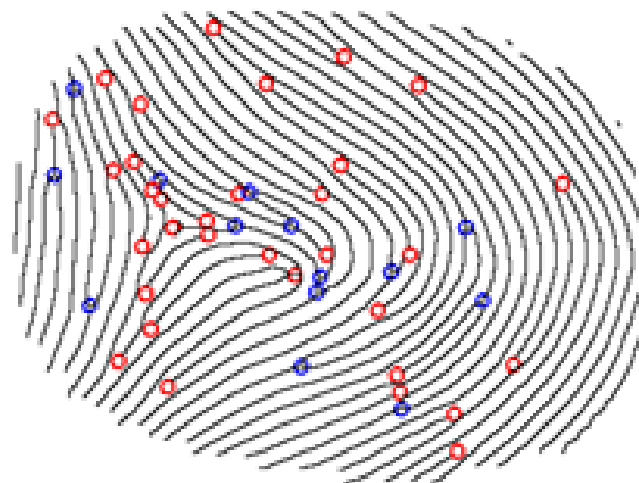
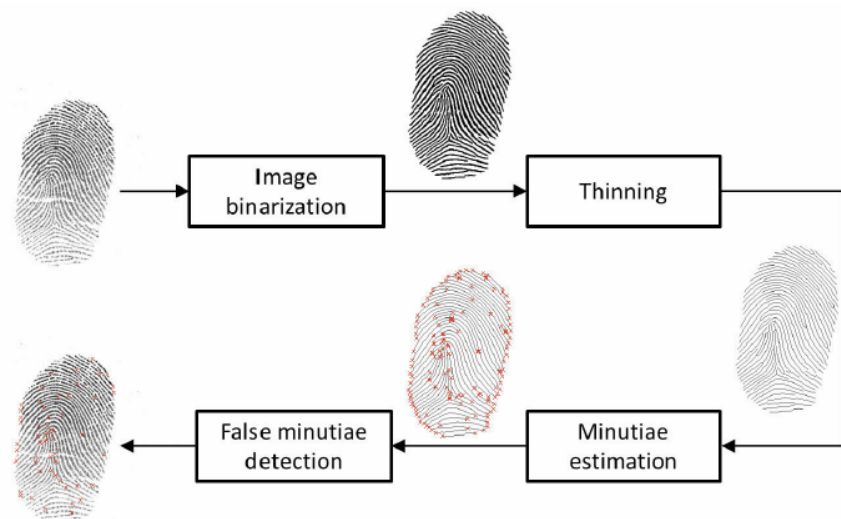
Valore della intensità dei pixel



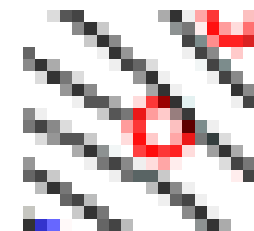
Interpolazione sinusoidale



Livello 2: Estrattore di minutiae più frequente



Esaminiamo 1 punto alla volta



Crossing Number 8 neighborhood of the pixel p

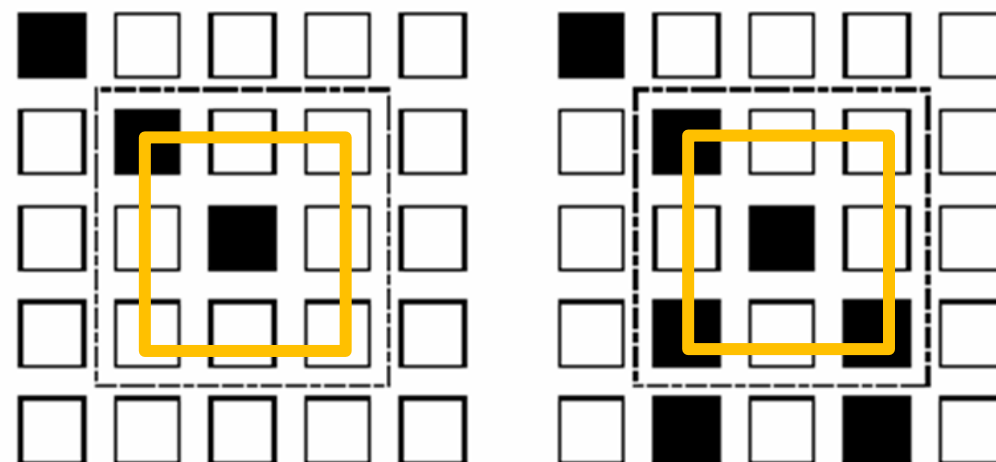
$$CN(p) = \frac{1}{2} \sum_{k=1}^8 |n_k - n_{((k+1) \bmod 8)}|$$

If $CN(p) = 1 \rightarrow$ FINE RIGA

If $CN(p) = 3 \rightarrow$ BIFORCAZIONE

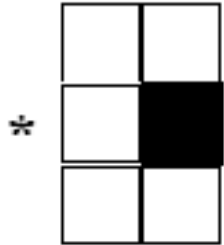
else $\rightarrow$ NESSUNA.

Esaminiamo il numero di transizioni nel percorso

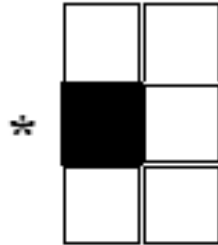


Livello 2: Esempi di pattern associati a fine riga e biforcazioni nelle immagini scheletrizzate

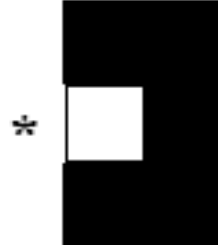
1. Ridge Ending
(appearing)



2. Ridge Ending
(disappearing)



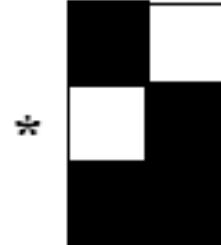
3. Bifurcation
(disappearing)



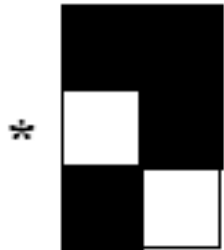
4. Bifurcation
(appearing)



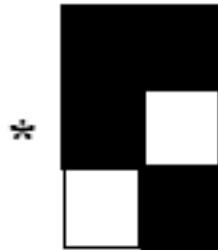
5. Bifurcation
(disappearing)



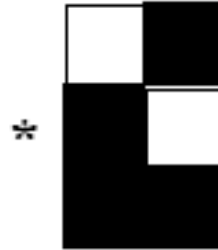
6. Bifurcation
(disappearing)



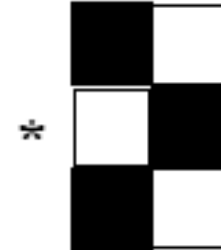
7. Bifurcation
(appearing)



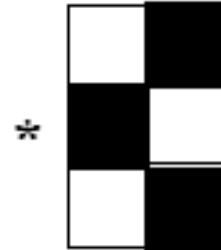
8. Bifurcation
(appearing)



9. Bifurcation
(disappearing)

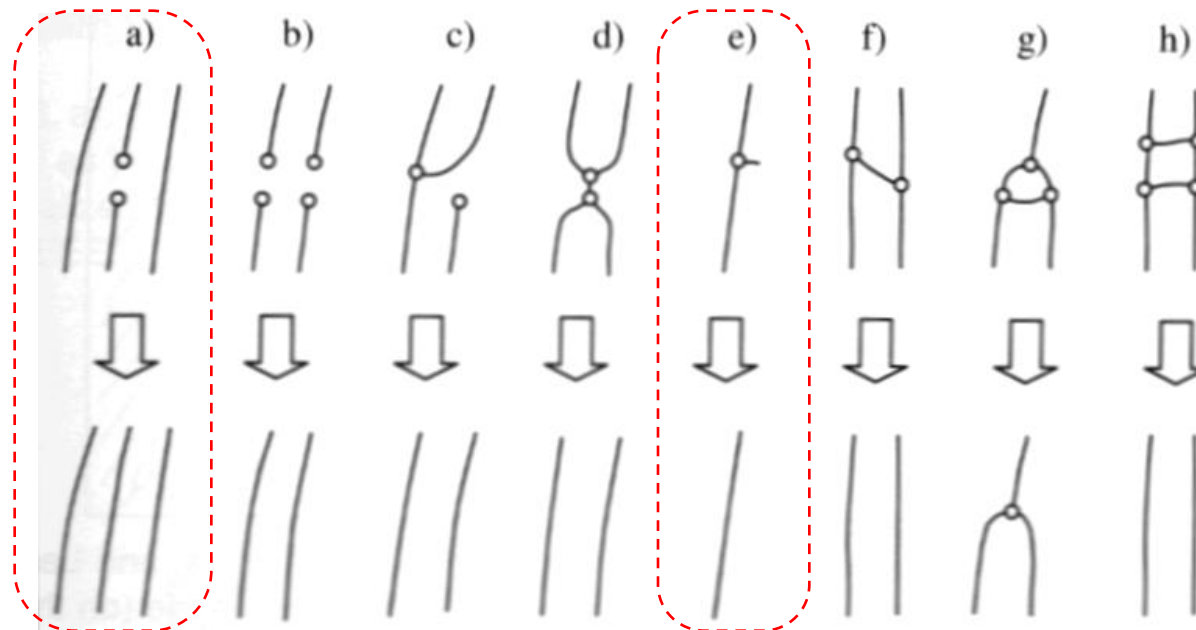


10. Bifurcation
(appearing)



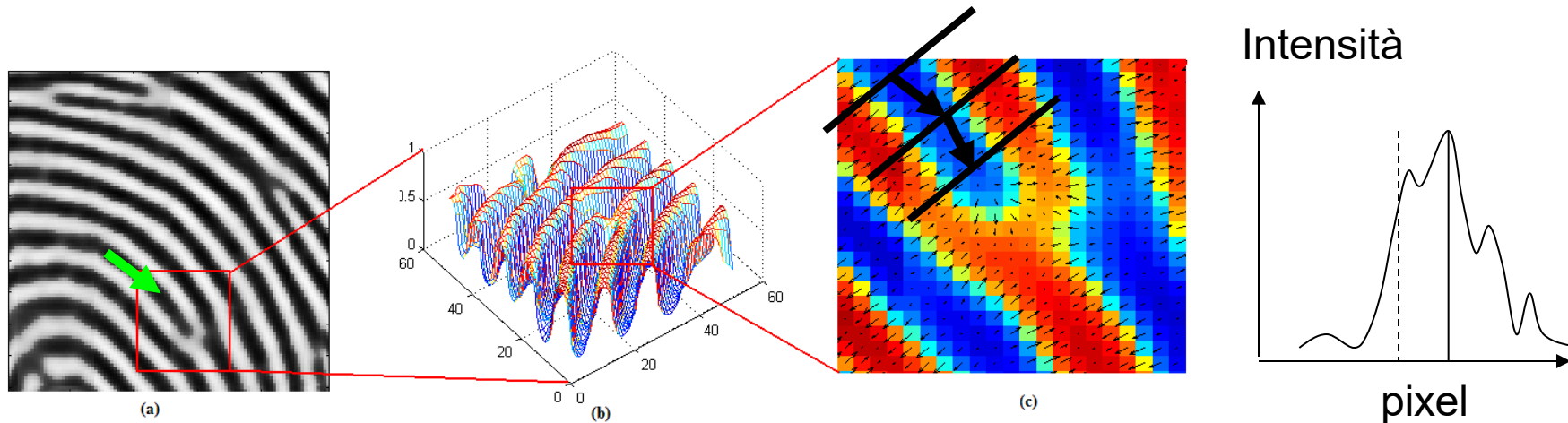
Structural post processing ⁽¹⁾

- I moduli di structural post processing sono tipicamente basati su regole
- Controllano le caratteristiche dello scheletro adiacente alla minugia candidata ad essere tenuta o scartata
- Le regole maggiormente usate sono queste 8



Direct Gray Scale Extraction

- Si sceglie un insieme di punti di partenza con una griglia
- Si segue ogni ridge dal punto di partenza fino a quando si trova un fine riga o una biforcazione
- Si usa una strategia di labellizzazione per evitare di seguire più volte lo stesso ridge



Matching livello 1: Filtri di Gabor

The even-symmetric Gabor filter has the general form

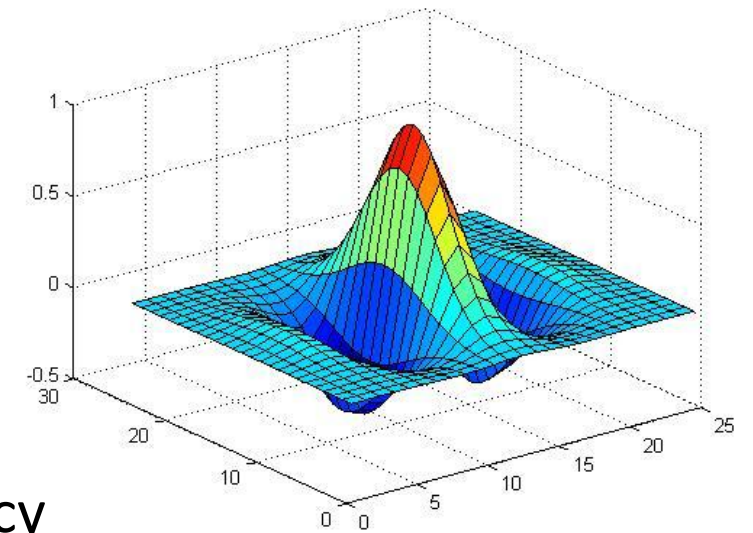
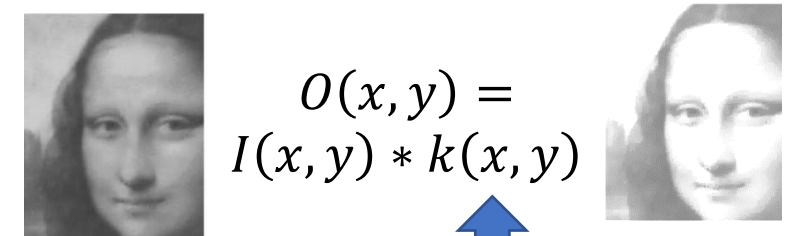
$$h(x, y : \phi, f) = \exp \left\{ -\frac{1}{2} \left[\frac{x_\phi^2}{\sigma_x^2} + \frac{y_\phi^2}{\sigma_y^2} \right] \right\} \cos(2\pi f x_\phi),$$

$$x_\phi = x \cos(\phi) + y \sin(\phi)$$

$$y_\phi = -x \sin(\phi) + y \cos(\phi)$$

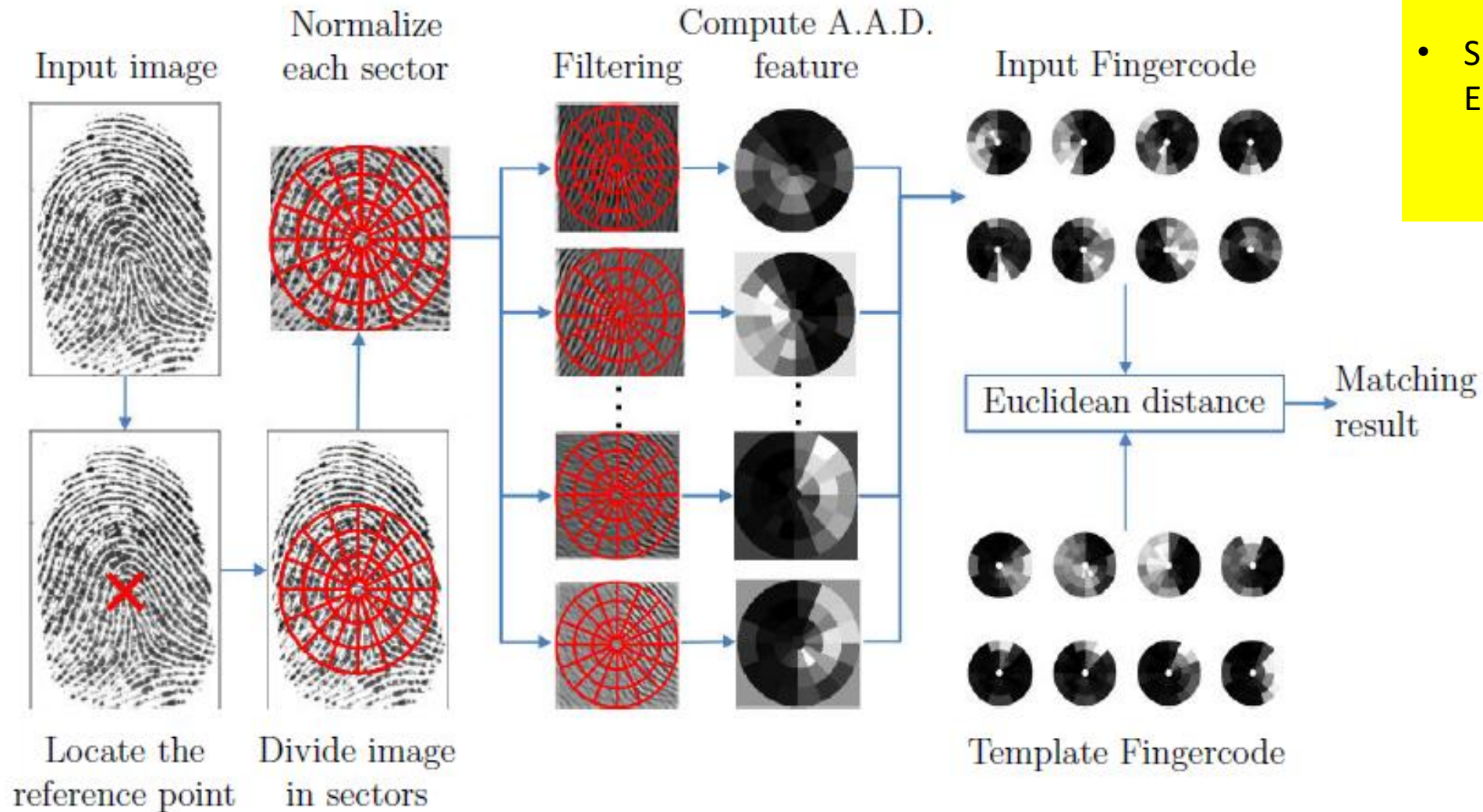
Rotazione
con angolo
parametrico

where Φ is the orientation of the Gabor filter, f is the frequency of a sinusoidal plane wave, and σ_x and σ_y are the space constants of the Gaussian envelope along x and y axes, respectively.



$k(x, y)$

Matching Livello 1: FingerCode



Il template FingerCode è importante perché

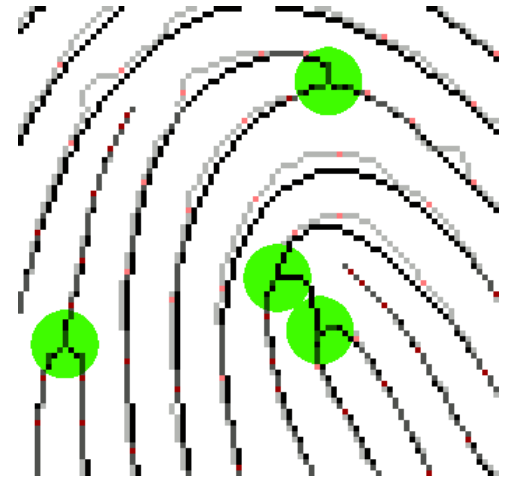
- di livello 1
 - funziona anche con basse risoluzioni
- Si può usare una metrica Euclidea
 - bassa complessità

Matching Livello 2 (base): Minutiae-based

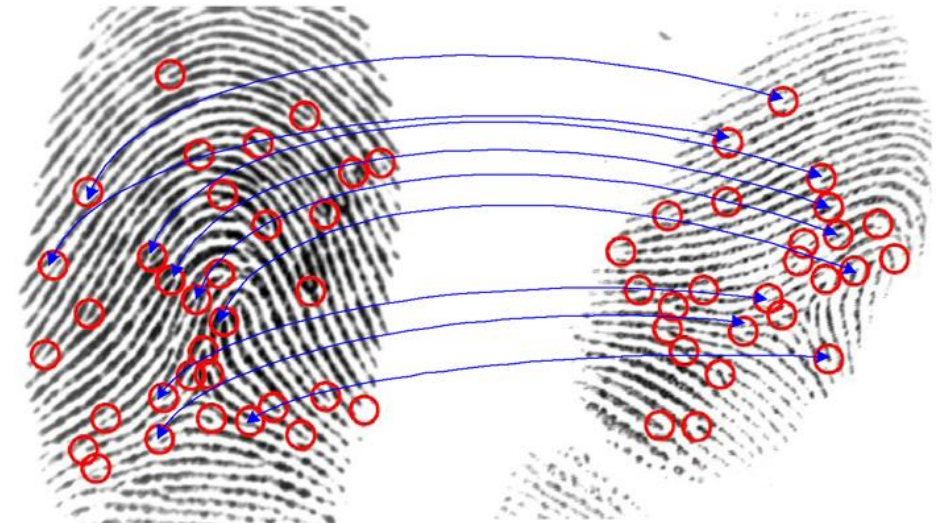
- Two minutiae are considered as correspondent if their spatial distance s_d and direction difference d_d are less than fixed thresholds

$$s_d(m'_j, m_i) = \sqrt{(x'_j - x_i)^2 + (y'_j - y_i)^2}$$

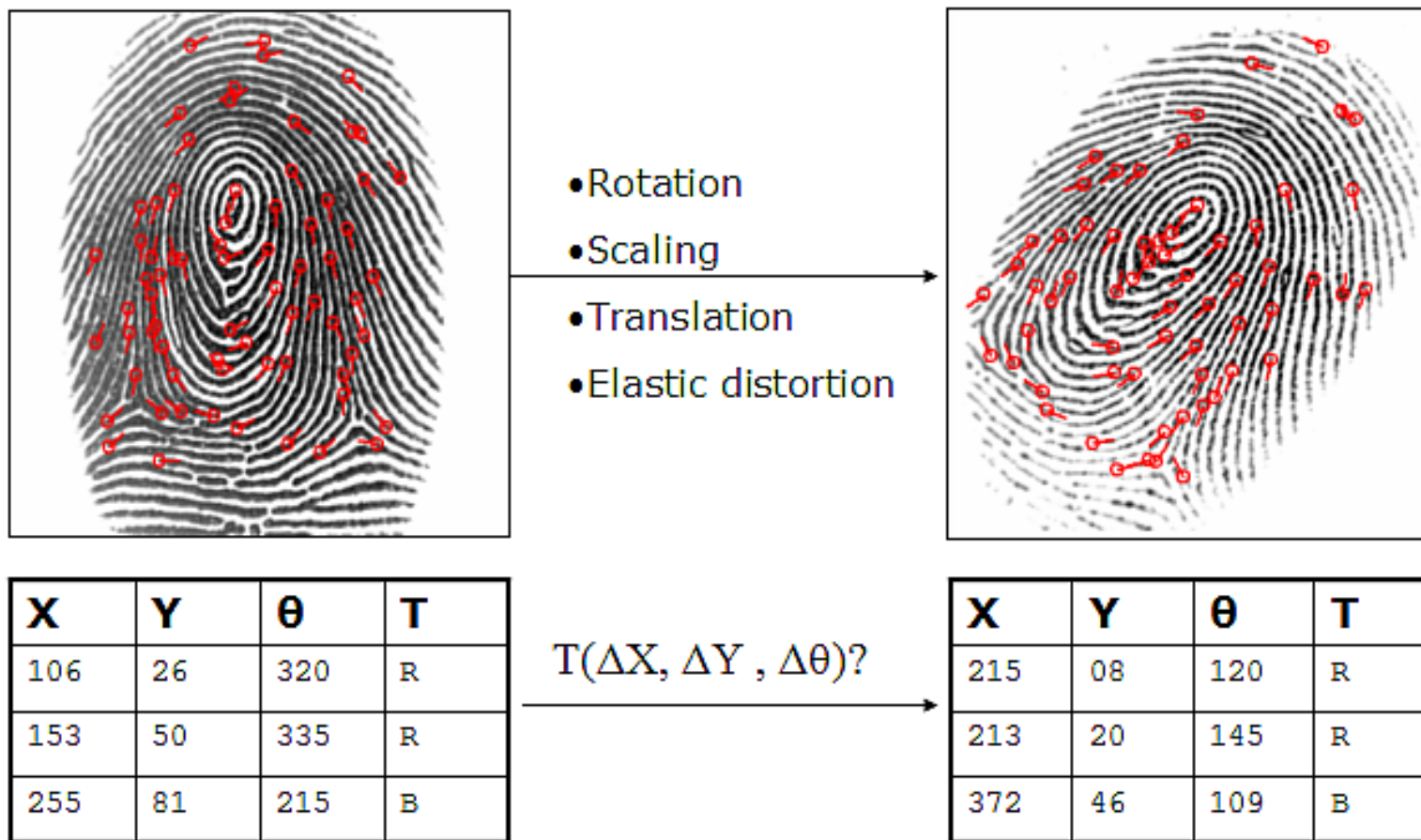
$$d_d(m'_j, m_i) = \min(|\theta'_j - \theta_i|, (360 - |\theta'_j - \theta_i|))$$



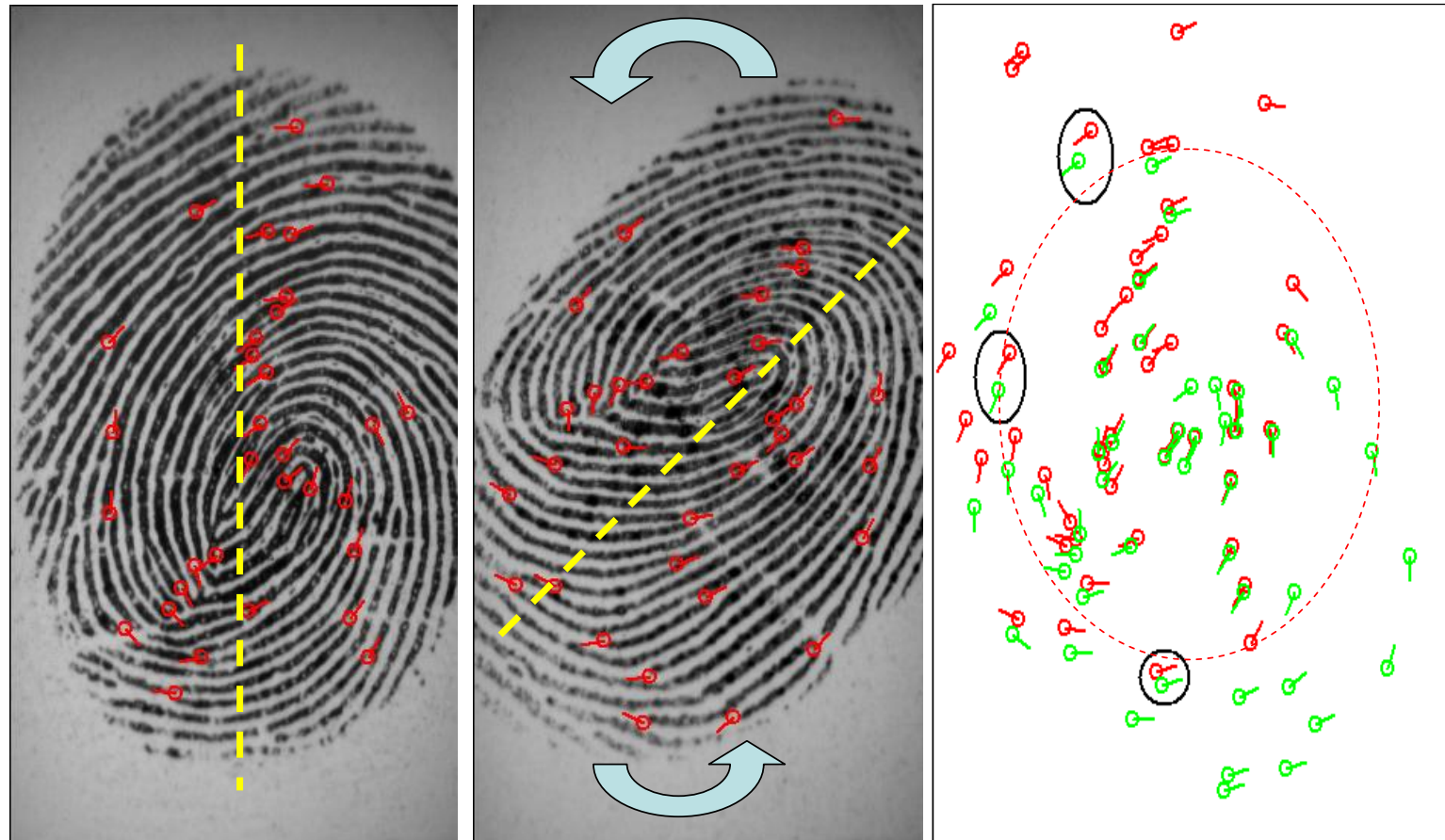
- Non-exact point pattern matching
 - Roto-translations
 - Non-linear distortions
 - False minutiae
 - Missed minutiae
 - Non constant number of minutiae



Matching Livello 2 classico (con angoli):

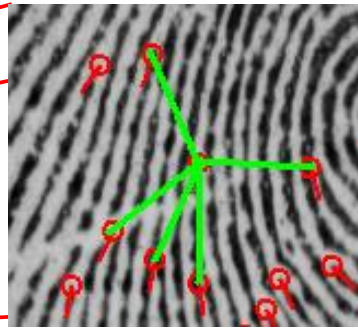
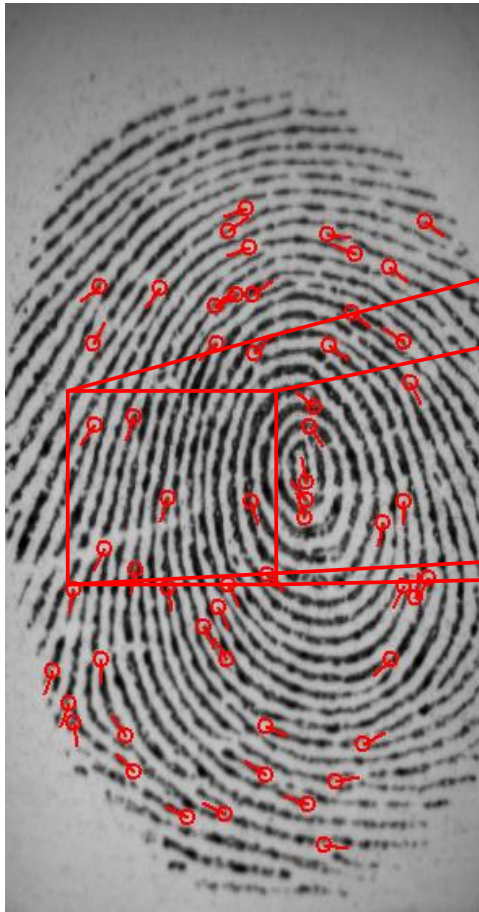


Matching con minutiae classico



Hybrid matching

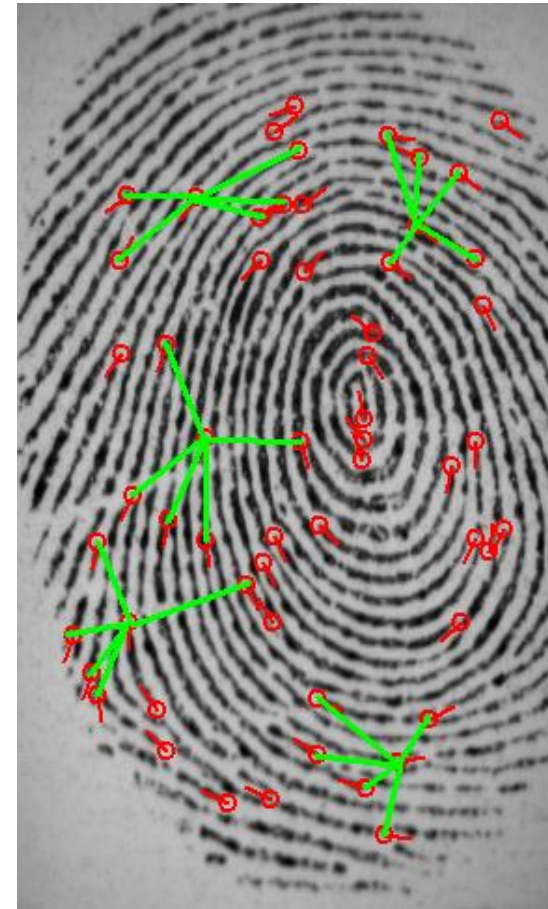
Input



Comparing subsets with:

- Minutiae types
- Angles
- Distances

Reference



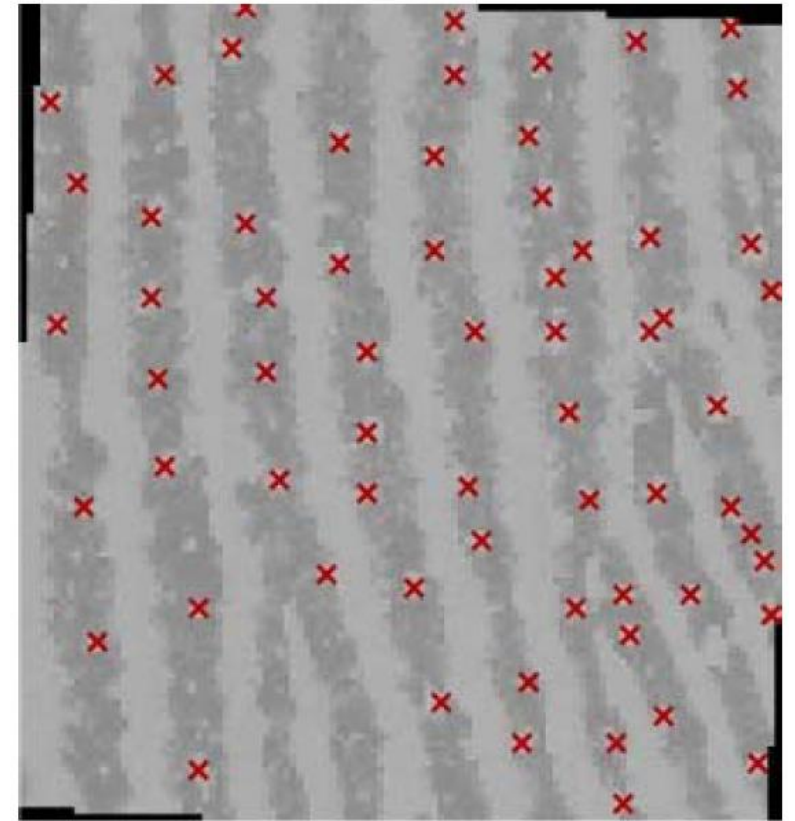
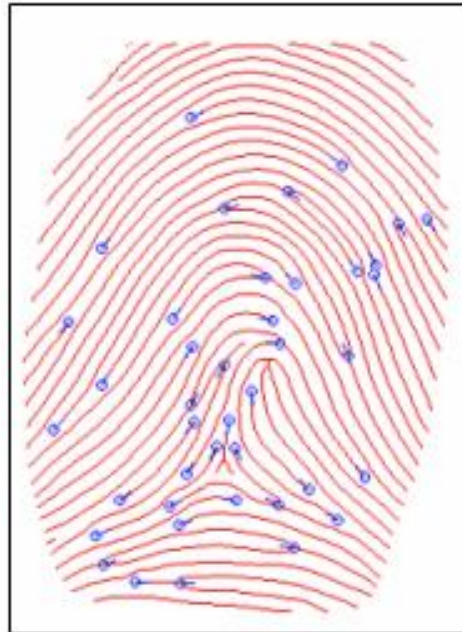
Esempio conclusivo

- Ecco alcune feature estratte sui 3 livelli



Livello I:
Mappa dei ridge

Livello II:
Minutiae

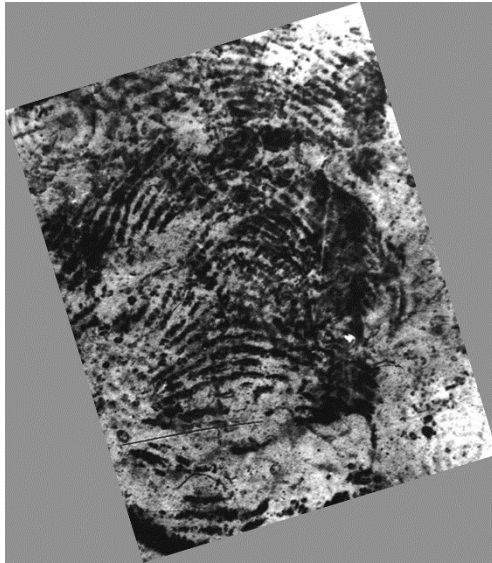


Livello III:
Pori

Dove è l'errore ?

In tutti questi casi ha fallito l'esaminatore
Le impronte sono diverse!

Latent (Matrid)



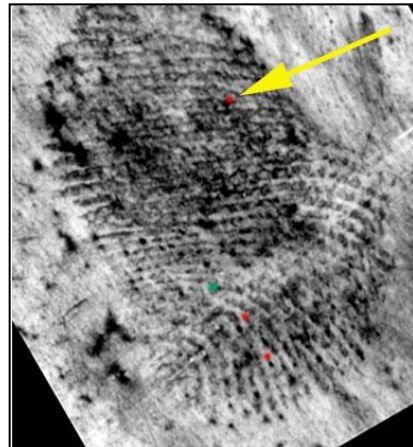
Vero attentatore
(Ouhnane Daoud)



Falso Positivo
(Brandon Mayfield)



Latent
(della scena
dell'omicidio
Marion Ross
Scozia)



Falso Positivo
(Shirley McKie)

SISTEMI AFIS E NAZIONALI per le IMPRONTE

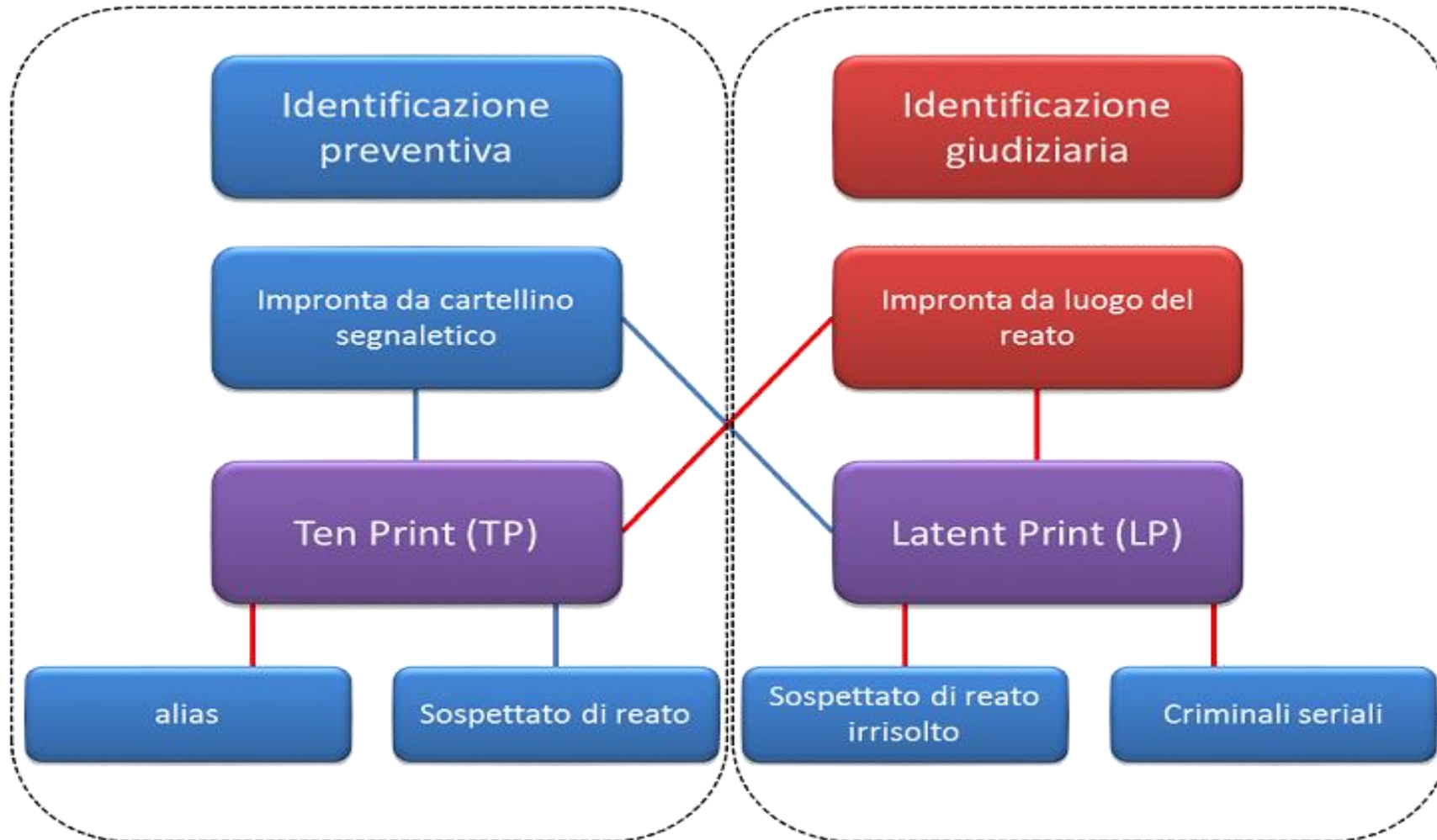
Il Sistema AFIS italiano



Progetto	• 1994
Inizio funzionamento	• 1995
Hardware	• Hewlett-Packard • Cogent System
Software AFIS	• Cogent System
Statistiche	• 12,5 milioni di cartellini • 7,5 milioni di individui memorizzati

Funzionamento

Impronte acquisite con un rapporto 1:1, ad una risoluzione di 500dpi, in scala di grigi
(per identificazione giudiziaria necessaria la presenza di un riferimento metrico)



Tecnologia sul mercato (1)

- Sistema della ditta M2SYS
- AFIS | ABIS (Automated Biometric Identification Systems)
 - storing, editing, processing, and searching biometric records and templates
 - multi-modal identification capabilities allows it to cross match a subject's fingerprint with a facial or iris modality.
 - FBI, NIST, ISO, ANSI certified
 - biometric matching server architecture supports 1:N and 1:1 matching systems and can search over 200 million irises or 100 million fingerprints per second.

Tecnologia sul mercato (2)



- Sistema della ditta DERMALOG
- AFIS | ABIS (Automated Biometric Identification Systems)
- The world's fastest matching: 129 million fingerprints within one second (on a standard Intel blade)
- Database with tenprints (TP), palmprints (PP) and latent prints (LP)
- Lowest FAR and FRR on the market
- Use of standard databases, such as Oracle
- Runs on all hardware platforms and common operating systems, such as Windows and/or Unix/Linux

Es. Tecnologia SDK della Innovatrics

- Alte prestazioni, integrabilità, standard (WSQ), piattaforme multiple
- Match on card
- Mobile, PC, Server, Embedded
- 99.98% Accuracy
- 1.040.000.000 impronte/s

	IDKit SDK 	ANSI & ISO SDK 	Segmentation SDK 	Match On Card 
Free evaluation	Available	Available	Available	Demo with Innovatrics cards
Maximum number of records	10k for PC and 500k for server edition	-	-	-
Standards compliance	• ISO/IEC 19794-2 • ISO/IEC 19794-4 • ANSI/INCITS 378	• ISO/IEC 19794-2 • ANSI/INCITS 378 • ILO SID • ISO/IEC 19794-4	-	• ISO/IEC 19794-2 • ISO/IEC 7816
Certifications and Benchmarks	• NIST PFT II • NIST FpVTE	• NIST MINEX • STQC	-	• NIST MINEX II
Quality Assurance	• Image Quality • Template Quality	• Image Quality	• NIST NFIQ • Image Quality • Fingerprint Placement Quality	-
Verification	• Innovatrics Proprietary Template • Innovatrics Compact Template	• ISO/IEC 19794-2 • ANSI/INCITS 378 • ILO SID • ISO/IEC 19794-2 CC	-	• ISO/IEC 19794-2 CC
Platforms / Operating system	• Windows • Linux • Android	• Windows • Linux • Android	• Windows • Linux • Android	-
Embedded Platforms	Available upon request	Available upon request	Available upon request	-
Use Cases	• Custom Build Applications for Fingerprint Verification and Identification • Local Fingerprint-Based Blacklists, Watch Lists and White Lists • Mobile Fingerprint Authentication and Identification	• PIV Programs: ANSI / ISO or ILO SID with Interoperable Templates	• Fingerprint Segmentation with Livescan Fingerprint Scanners • Fingerprint Enrollment Solutions • Fingerprint Image Quality • Finger Placement Score • NFIQ Calculation	• PIV programs • Access Control • Two Factor Authentication

Trend nelle applicazioni



Servers



Dispositivi
embedded



Contactless



Mobile



In sintesi

Impronte a contatto

- Biologia e caratteristiche
- Tipi di sensori a stato solido e ultrasuoni
- Sistemi commerciali e prestazioni
- Tipi di impronte (classificazione)
- Analisi sui 3 livelli della impronta
- Estrazione delle minutiae (metodo classico e direct gray scale)
- Matching delle impronte
- Esempio di un sistema AFIS
- Trends

L3

